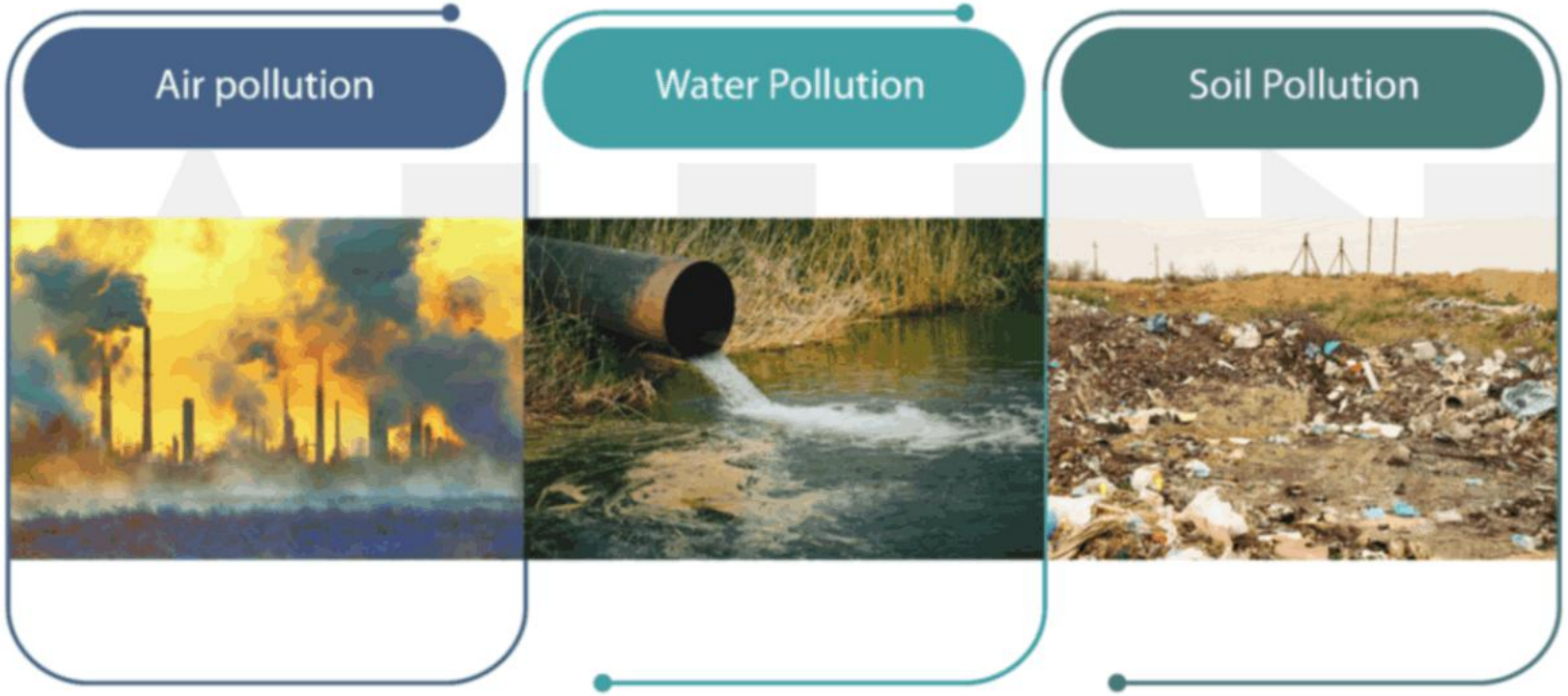
Part-2: Organic chemicals in the Environment

What is Environmental Chemistry?

- ***Environmental Chemistry is the branch of Chemistry that focuses on understanding the chemical processes occurring in the environment, including air, water, soil, and living organisms***.
- Introduction to environmental Chemistry involves studying the sources, reactions, transport, effects, and fate of chemical species in natural and polluted environments.
- Pollution is the introduction of contaminants into the natural environment, causing adverse changes. Here we will learn some major types of Pollution under the topics of environmental Chemistry.



What is green chemistry?

"Green chemistry is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances. Green chemistry applies across the life cycle of a chemical product, including its design, manufacture, use, and ultimate disposal." - U.S. Environmental Protection Agency (EPA).

What is green engineering?

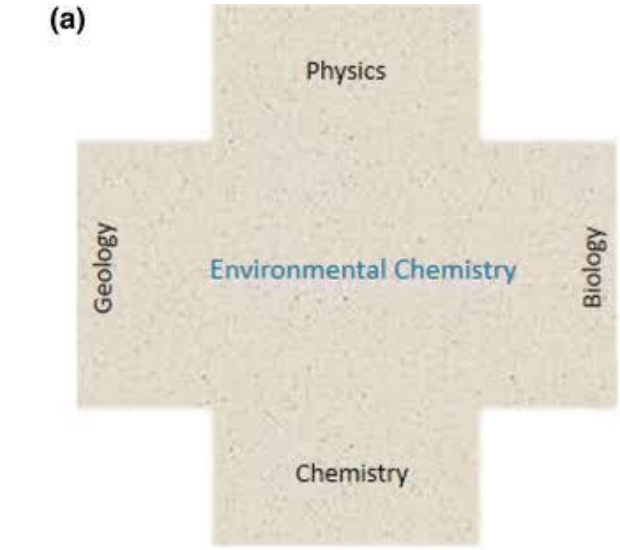
"Green engineering is the design, commercialization, and use of processes and products in a way that reduces pollution, promotes sustainability, and minimizes risk to human health and the environment without sacrificing economic viability and efficiency."

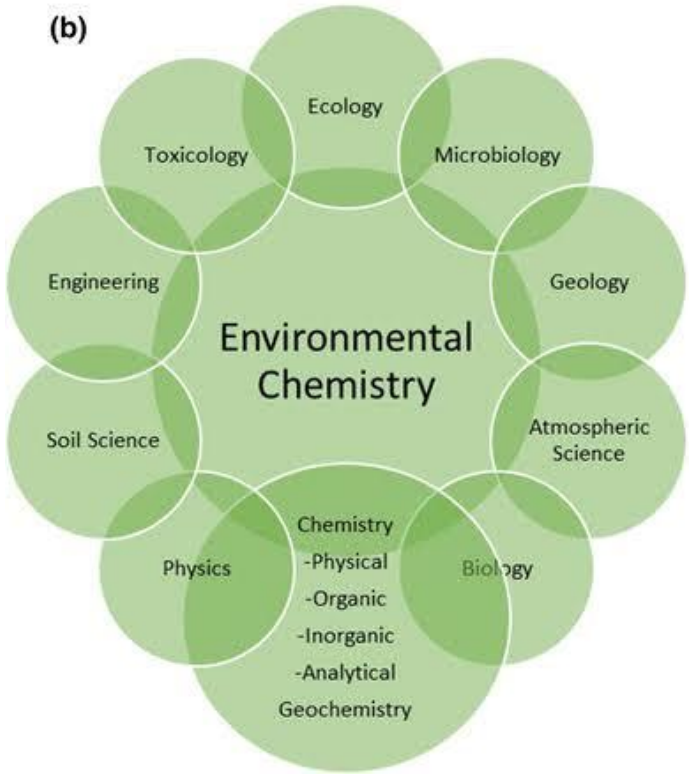
What is Sustainability: Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

Sustainability and sustainable development are closely related but have distinct meanings. Sustainability is a broad concept focused on the long-term ability to maintain balance and avoid resource depletion, while sustainable development refers to the processes and actions taken to achieve this balance, emphasizing development that meets current needs without compromising the ability of future generations to meet their own.

Environmental Chemistry Applications

1. **Monitoring Environmental Quality:** Assessing air, water, and soil conditions to identify and quantify pollutants.
2. **Pollution Remediation:** Developing methods to clean contaminated air, water, and soil, including bioremediation and phytoremediation.
3. **Public Health Protection:** Evaluating chemical exposure risks to safeguard human health.
4. **Informing Policy and Regulation:** Providing scientific data to shape environmental laws and standards.
5. **Advancing Sustainable Practices:** Promoting green Chemistry and resource recovery to minimize environmental impact.
6. **Addressing Climate Change:** Investigating and mitigating the effects of greenhouse gases.





Ref: https://link.springer.com/article/10.1007/s10311-016-0601-3

Atmospheric Pollution

Atmospheric pollution in the troposphere, the lowest layer of Earth's atmosphere, consists of undesirable solid and gaseous particles that adversely affect air quality, health, and the environment. Here's a concise summary of the major pollutants found in the troposphere:

Gaseous Air Pollutants

Oxides of Sulfur (e.g., SO_2): Produced from burning sulfur-containing fuels, leading to respiratory issues and environmental damage.

Oxides of Nitrogen (e.g., NO, NO_2): Formed during high-temperature combustion processes, contributing to smog, respiratory issues, and ecosystem damage.

Hydrocarbons: Result from incomplete combustion in vehicles, associated with cancer risks and environmental stress.

Carbon Monoxide (CO): A colorless, odorless gas from incomplete combustion, highly toxic due to its ability to impair oxygen transport in the bloodstream.

Carbon Dioxide (CO_2): Emitted from burning fossil fuels and other organic materials, a key greenhouse gas contributing to global warming.

What are Organic Chemicals?

Organic chemicals contain carbon atoms. Natural ones include sugars and plant oils. Synthetic (man-made) ones include plastics and drugs. Synthetic ones often cause pollution when thrown away

Types of Organic Pollutants

Volatile Organic Compounds (VOCs):

These liquids evaporate quickly into the air. Examples are benzene, paint thinners, and cleaning fluids.

Persistent Organic Pollutants (POPs):

These toxic chemicals stay in the environment for a very long time. They include old pesticides (like DDT) and industrial chemicals.

Pesticides

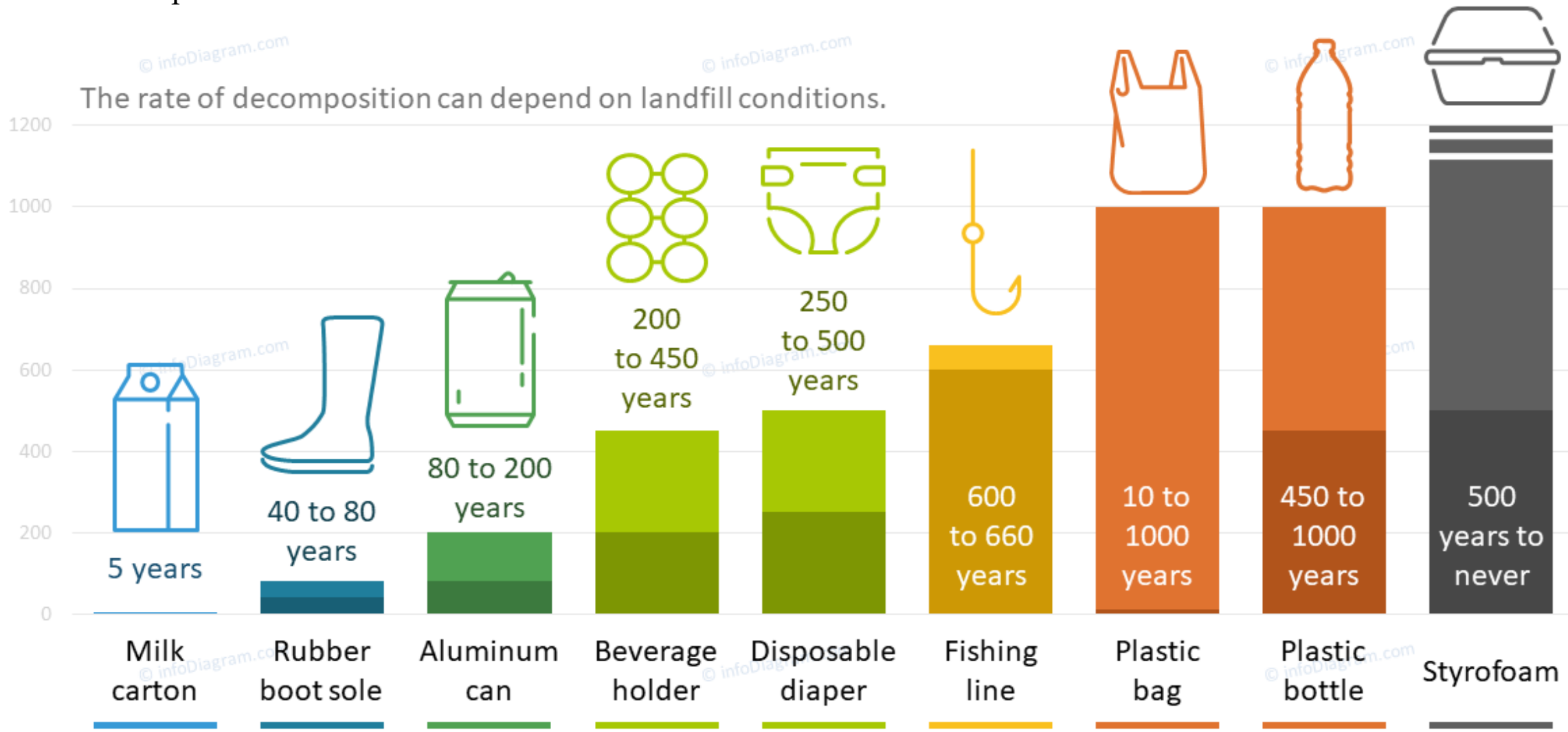
Depletion of ozone ayer

Plastic pollution

It is estimated that about 8 million metric tons of plastic waste enters the ocean each year

How Long Does Plastic Take to Decompose?

Plastics can take 20 to 500 years to decompose, depending on the material and structure.
Here are different plastics ranked.



Ref: https://www.infodiagram.com/slides/time-decompose-plastics/

Common examples of single-use plastic products

Polyethylene terephthalate [Water bottles]

High-density polyethylene [Milk bottles]

Low-density polyethylene [Packaging film]

Polyethylene Polypropylene [Grocery bags]

Polystyrene [Cutlery]

Polyvinyl chloride [Plumbing pipes]

Polycarbonate [Construction]

Polyurethane [Foam, sponge]

FILTER COFFEE

How much waste does the world generate daily

Amount in kg per capita per day

Electronic Waste

0.02 Kg

Industrial Waste

12.73 Kg

Medical Waste

0.25 Kg

Kg per capita per day shows how many kilograms of waste one person generates on average each day.

Agricultural Waste

3.35 Kg

Hazardous Waste

0.32 Kg

Municipal Solid Waste

0.74 Kg

Construction & Demolition Waste

1.68 Kg

Source: Global Analysis on Crimes that Affect the Environment

LINEAR ECONOMY

Take

Make

Use

Waste

RECYCLING ECONOMY

Take

Make

Recycle

Use

Waste

CIRCULAR ECONOMY

Take

Make

Recycle

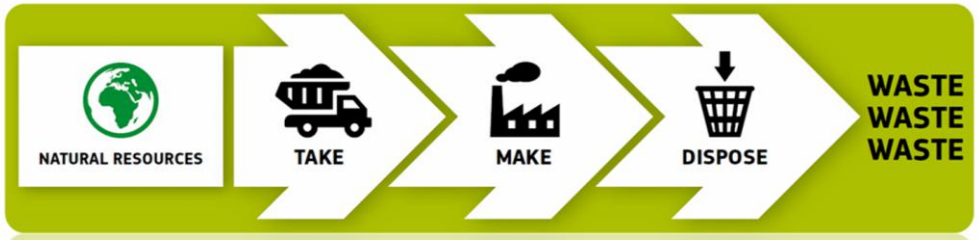
Return

Use

Repair

Reuse

Linear Economy



Definition: A linear economy follows the “take-make-dispose” model.

Raw materials are extracted, converted into products, and discarded as waste after use.

Characteristics:

•**Resource Extraction**: Natural resources are extracted to make products.

•**Production**: Raw materials are converted into goods.

•**Usage**: Consumers use the products for a certain period.

•**Waste**: At the end of their life, products are often discarded as waste, leading to landfills or incineration.

Disadvantages:

•**High Environmental Impact**: Depletion of natural resources and increased waste production.

•**Short Product Lifespan**: Products are often designed for single-use or short lifespan.

Recycling

Definition: Recycling is the process of converting used materials into new products to reduce waste and conserve resources.

Characteristics:

- **Waste Collection**: Materials such as paper, glass, metal, and plastic are collecte
- **Processing**: These materials are sorted, cleaned, and processed into raw materia
- **Production**: Recycled raw materials are used to make new products.

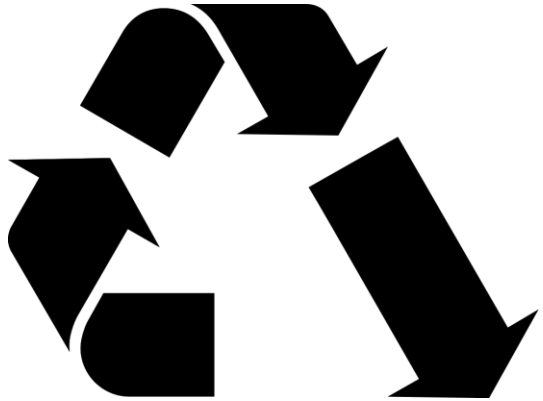
Advantages:

- **Waste Reduction**: Fewer materials end up in landfills or are incinerated.
- **Resource Conservation**: Less need for new raw materials, reducing the pressure on natural resources.

Limitations:

- **Quality Loss**: Recycled materials may degrade in quality, limiting their usefulness for new products.
- **Energy Consumption**: Recycling processes require energy, which can have an environmental impact

Downcycling



Definition: Downcycling is a form of recycling where the recycled material is converted into products of lower quality and functionality than the original material.

Characteristics:

- **Quality Reduction**: Materials lose quality with each recycling cycle.
- **Limited Applications**: Recycled materials are often used for products with lower performance requirements.
- **Example**: Artificial grass can often only be converted into lower-quality products like filler material or secondary construction materials after use.

Advantages:

- **Waste Reduction**: Although quality decreases, the material stays in use longer and out of the waste stream.
- **Resource Conservation**: Better than disposal, but not as effective as high-quality reuse.

Limitations:

- **Quality Loss**: The material is no longer suitable for high-quality applications, reducing its circular value.
- **Temporary Solution**: Eventually, the material often ends up as waste.

Circular Economy

Definition: A circular economy is a regenerative system where resources, materials, and products are kept in use for as long as possible, reused, and recycled to minimize waste and environmental impact.

Characteristics:

•**Design for Sustainability**: Products are designed for longevity, reuse, and easy recycling.

•**Closed Loop**: Materials circulate within the system without losing value.

•**Reuse and Repair**: Products and components are reused and repaired instead of discarded.

Advantages:

•**Minimal Waste Production**: By reusing and recycling materials, the amount of waste is greatly reduced.

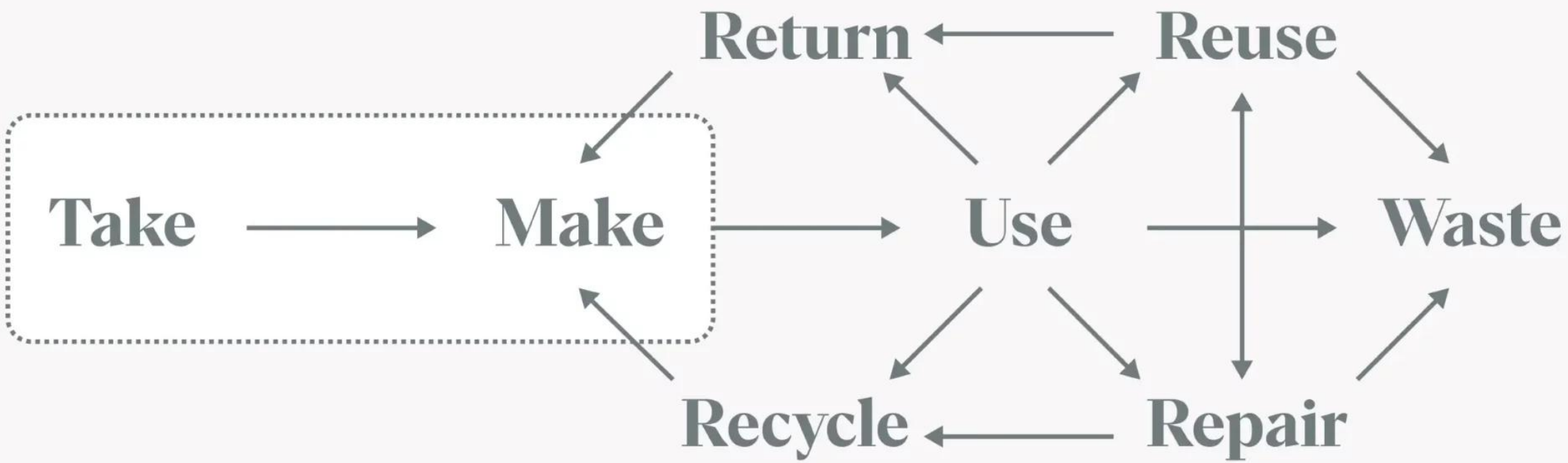
•**Value Retention**: Materials retain their value and utility within the system, which is economically beneficial.

•**Sustainable Resource Use**: Reduces the need for new raw materials and minimizes environmental impact.

Challenges:

•**Infrastructure and Systems**: Requires new infrastructures and economic models.

•**Behavioral Change**: Requires a change in how businesses and consumers handle products and materials.


Return
Reuse
Take
Make
Use
Waste
Recycle
Repair

Reduce Environmental Footprint
Green products
non-toxic, long-life, recyclable
Cleaner production
using fewer resources
Generate Increased Income
Circular Economy
Recycle waste, reuse resources
Better service
to extend lifespan
Minimize Waste
Collect at end-of-life, remanufacture
Reduce Resource Dependency

WHAT 1 TON OF FOOD WASTE CAN DO

Waste Innovation Stories

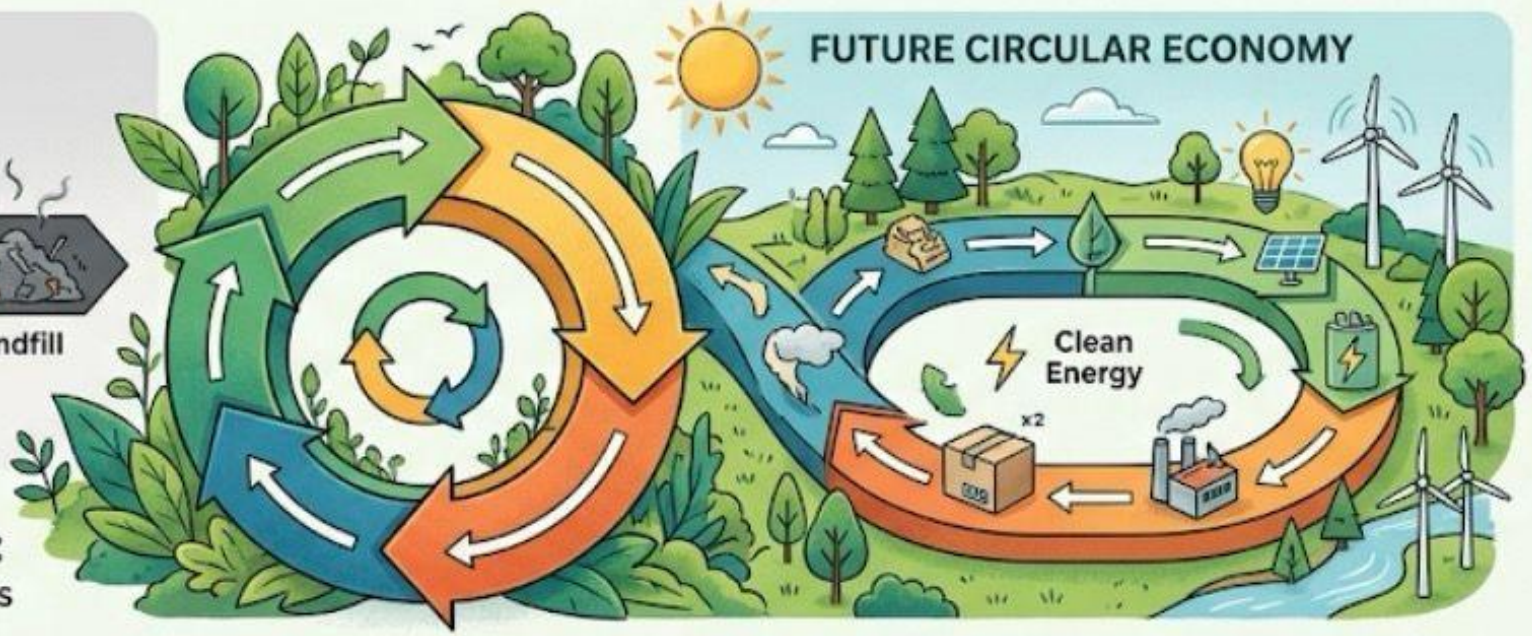
ENERGY	WATER	COMPOST	EMISSIONS
Generates enough biogas to power 8–12 homes for a day	Saves the equivalent of 15,000–25,000 liters of embedded water (depending on the food type)	Produces around 400–600 kg of nutrient-rich compost	Avoids 0.5–1.0 tons of CO_2e emissions when diverted from landfill

Food waste isn't garbage – it's lost energy, water and life. Let's recover it.

THE FUTURE IS CIRCULAR, NOT LINEAR.

OLD LINEAR ECONOMY
(Take-Make-Waste)

FUTURE CIRCULAR ECONOMY



PRINCIPLES OF CIRCULARITY:
A Practical Roadmap for Smarter Systems

1. **REFUSE –** Hand rejecting unnecessary shopping bags.
2. **RETHINK –** Lightbulb with blueprint designs.
3. **REDUCE –** Decreasing arrows and footprints.
4. **REUSE –** Persons swapping items, a shopping tote.
5. **REPAIR –** Tools fixing a toaster and bicycle.
6. **REFURBISH –** Restoring furniture and electronics.
7. **REMANUFACTURE** Assembly line rebuilding parts. and inotnores.
8. **REPURPOSE –** Materials turning into new objects (e.g., tires to swings).
9. **RECYCLE –** Sorting bins with the recycling arrows symbol.

THE GOAL: Design Waste Out of the System Entirely.

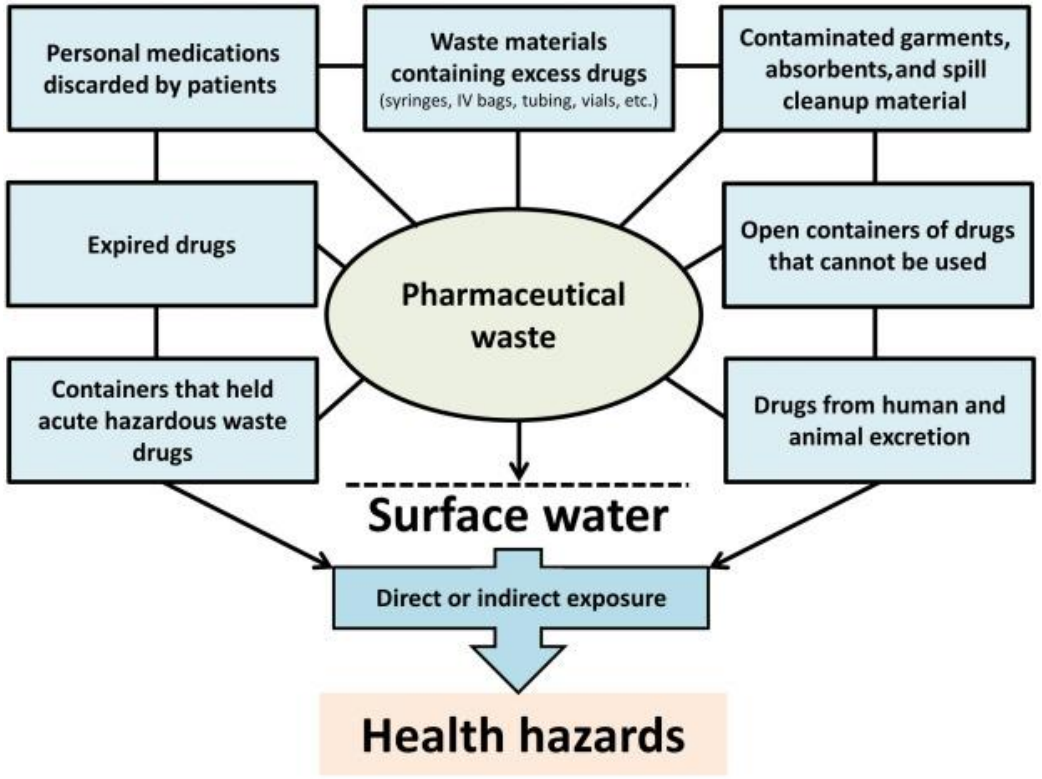
Circularity is reshaping how value is created and preserved.

The future economy will be defined by how long resources are kept in use.

Pharmaceutical wastes

Pharmaceutical wastes are potentially generated through a wide variety of activities in the health care system

Pharmaceutical manufacturing waste includes expired products, contaminated materials, hazardous pharmaceutical substances, and unused medicines generated during GMP manufacturing operations. This category includes expired drugs, damaged medications, unused medications, and medicines unsuitable for their intended use. It also includes partially filled or empty medicine containers. **Pharmaceutical waste** is generated across various healthcare settings, from **pharmacies** to hospitals, clinics, and even households.



Identifying and sorting pharmaceutical waste is crucial for proper management and safe disposal. Some common types of pharmaceutical waste include:

Hazardous medicines: Medicines that pose significant health and environmental risks if not managed correctly (e.g. Highly Potent Active Pharmaceutical Ingredients (HPAPIs))

Non-hazardous medicines: Medicines that do not exhibit hazardous characteristics but still require special handling to prevent misuse or contamination.

Controlled drugs: These are medications regulated under specific legislation due to their potential for abuse or addiction.

Waste containers: Empty or partially filled medicine containers that may have been contaminated during use.

Pharmaceutical wastes

Key Principles of Pharmaceutical Waste Disposal

Pharmaceutical waste disposal in England follows principles aimed at minimizing environmental impact. These include:

Reduce: Minimizing the quantity of waste generated.

Reuse: Finding ways to reuse medications and containers where possible.

Reprocessed: Utilizing technologies that allow pharmaceuticals to be repurposed safely.

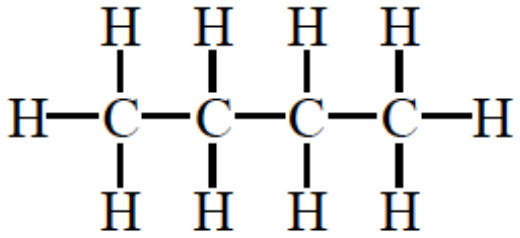
Renewable: Incorporating renewable resources in pharmaceutical production to lessen waste.

Recyclable: Identifying components of waste that can be recycled.

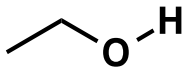
Organic Chemicals in the Environment

Organic compounds: Compounds that mainly contain carbon and hydrogen atoms

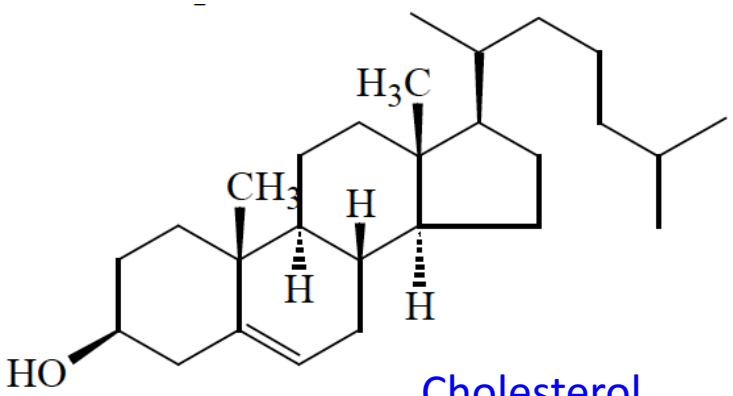
Ex.



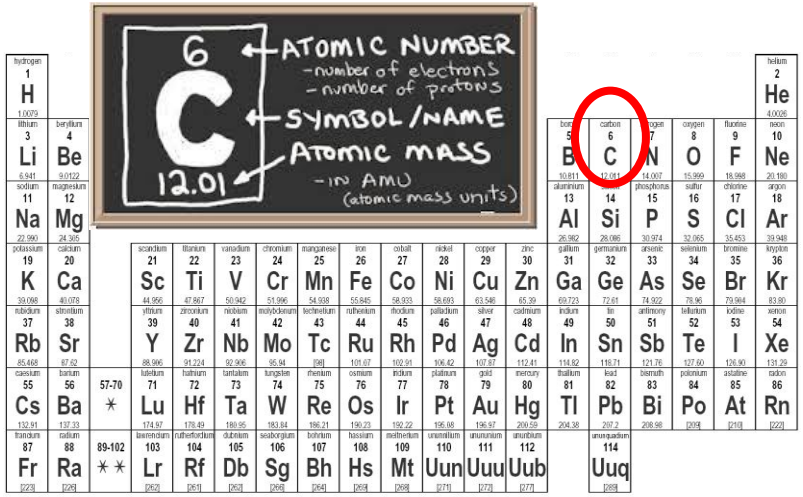
n-butane



Ethanol



Cholesterol



Carbon exhibits catenation property

Carbon is the basic building block for all forms of life on Earth

Glucose: $C_6H_{12}O$

DNA

Amino acids

Proteins

Medicines

Plastics

Pesticides

The Many Colors of Carbon

Greenonomics

Soot particles from incomplete combustion absorb sunlight intensely, warming air rapidly and accelerating ice melt.

Organic aerosols from biomass burning absorb light, influencing atmospheric heating, clouds, visibility and regional climate.

Carbon stored within mangroves, seagrasses and saltmarshes sequesters emissions long-term, stabilizing coasts and climates globally.

Carbon captured by forests, grasslands, wetlands, and soils supports biodiversity, resilience and balanced carbon cycles.

Biological material on snow and ice darkens surfaces, lowering albedo and accelerating melt rates significantly.

Emissions from fossil-fueled industries, power, transport and manufacturing drive warming and demand rapid decarbonization globally.

Organic Chemicals in the Environment

- Thousands of different organic chemicals are synthesized each year for use as cosmetics, insecticides, detergents, and plastics
- Some of them are not adequately tested for toxicity before being put on the market
- Many of these chemicals persist in the environment for long periods of time

Persistent organic pollutants (POPs)

Organic compounds that are resistant to degradation through chemical, biological and photolytic processes.

	Half-Life of Chemical		
Media	**Not Persistent**	**Persistent**	**Highly Persistent**
Water	< 2 months	≥ 2 months	> 6 months
Soil	< 2 months	≥ 2 months	> 6 months
Air	≤ 2 days		> 2 days
Sediment	< 2 months	≥ 2 months	> 6 months

Organic Chemicals in the Environment

Persistent organic pollutants (POPs)

- POPs can enter into water and food chains and can cause serious health and environmental problems
- POPs can go from one place to other via wind and water
- POPs generated in one country can affect the people and wildlife in other countries even though they are very far from each other
- Some POPs evaporate from water or land surfaces into the air, then return to Earth in snow, rain, or mist

Organic Chemicals in the Environment

Insecticides: Substances used to kill insects

Pesticides: Substances used to kill pests including weeds

Herbicides: Pesticide used to kill unwanted plants

Fungicides: Substances used to kill fungus

Approximately 80,000 synthetic chemicals are on the market today and most of them have never been tested for toxicity

Organic Chemicals in the Environment

- POPs contamination is also found in arctic regions, which are thousands of miles from anywhere the POPs are manufactured or used

Stockholm Convention

In 2001 **"dirty dozen" POPs**

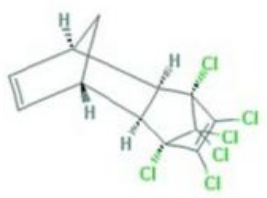
- Out of 12 chemicals, 10 were produced by industry for a purpose
- 9 were produced as **insecticides or fungicides**
- Two are unintentionally produced in combustion processes

Organic Chemicals in the Environment

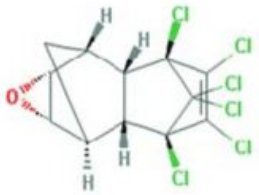
The 12 Key POPs—the Dirty Dozen

POP	Use
Aldrin	crop insecticide (corn, cotton)
Chlordane	crop insecticide (vegetables, citrus, cotton, potatoes)
DDT	crop insecticide (cotton)
Dieldrin	crop insecticide (cotton, corn)
Endrin	crop insecticide (cotton, grains)
Heptachlor	insecticide (termites and soil insects)
Hexa-chlorobenzene	fungicide for seed treatment
Mirex	insecticide (termites, fire ants)
Toxaphene	insecticide (livestock and crops)
PCBs	industrial chemical (heat exchange fluid for electrical transformers, paint and plastic additive)
Dioxins	unintentionally produced during combustion
Furans	unintentionally produced during combustion

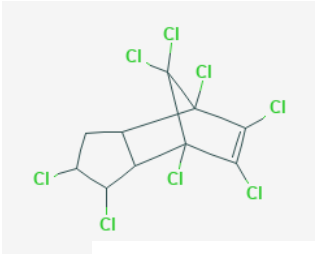
Organic Chemicals in the Environment



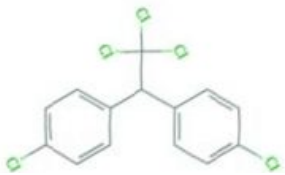
Aldrin



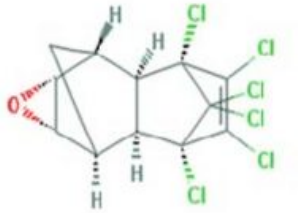
Dieldrin



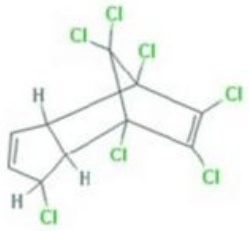
Chlordane



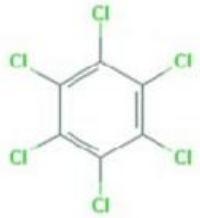
DDT
(Dichlorodiphenyltrichloroethane)



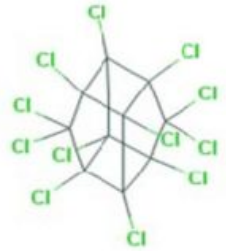
Endrin



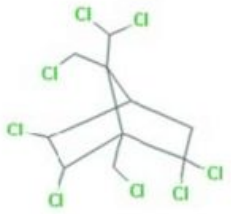
Heptachlor



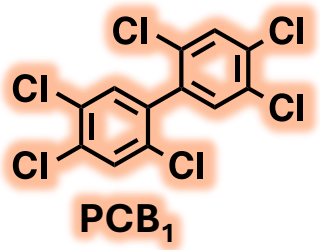
HCB



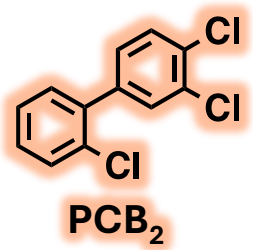
Mirex



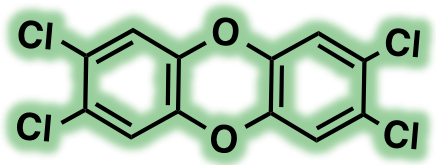
Toxaphene



PCB_1

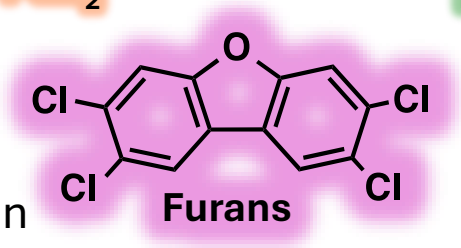


PCB_2



Dioxins

Polychlorinated compounds



Furans

2,3,7,8-tetrachlorodibenzo[*b*,*d*]furan

Synthetic Organic Chemicals:

Synthetic organic chemicals, often shortened to SOCs in drinking-water regulation, are carbon-containing chemicals produced through human-controlled chemical manufacturing or synthesis.

They are an enormous and diverse group.

Depending on the context, synthetic organic chemicals can include:

Pesticides and herbicides

Industrial solvents

Plasticizers

PFAS

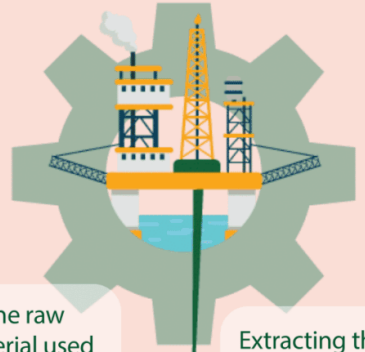
Polymers

Pharmaceuticals

Flame retardants

Many other industrial and consumer-product chemicals

Synthetic fibres VS Natural fibres



The raw material used for most synthetic textiles is oil.

Extracting this non-renewable fossil resource is highly energy-intensive & results in high CO_2 emissions.

Synthetic fibres take between 20 and 200 years to biodegrade: a challenge for waste management.

Synthetic textiles shed microplastics – when used, when washed, and after they have been disposed of.

Each year, between 200,000 and 500,000 tonnes of microplastics from textiles enter the marine environment, constituting a major environmental and health concern.

The production of natural fibers offers a livelihood to millions of farmers worldwide.

These fibres come from nature and are renewable materials

Ideally, fibres are organic and the environment & animal welfare is respected.

Organic textile standards make sure that organic natural fibres are further processed into clothes with responsibility and care for the environment and textile workers, and no hazardous chemicals are used in processing.

In chemistry, an **organic compound** generally refers to a carbon-based chemical structure.

A synthetic organic chemical is one produced through deliberate chemical synthesis rather than occurring solely through natural processes.

That definition covers a huge range of substances with very different properties.

Category	Examples	Common Uses or Sources
Pesticides	Atrazine, alachlor and other compounds	Agriculture and pest control
Industrial solvents	TCE, PCE	Degreasing, manufacturing, dry cleaning
Plasticizers	Phthalates such as DEHP	Flexible plastics and consumer products
PFAS	PFOA, PFOS and many others	Industrial processes and water-, grease- or heat-resistant products
Persistent organochlorines	PCBs	Historical electrical and industrial applications
Polymers	Polyethylene, polypropylene, PVC, nylon	Packaging, textiles and manufactured goods

Not all synthetic organic chemicals are drinking-water contaminants. Large polymers, such as intact polyethylene, behave very differently from small, dissolved chemicals such as TCE, atrazine, or certain PFAS.

Does “Organic” Mean Natural or Safe?

No.

In chemistry, **organic** does not mean the same thing as “organic food” or “natural.”

Many naturally occurring chemicals are organic compounds, and many manufactured chemicals are as well.

For example:

- Ethanol can be produced naturally through fermentation or industrially.
- Many pharmaceuticals are synthetic organic compounds.
- Some naturally occurring organic chemicals are toxic.
- Some synthetic organic chemicals have relatively low toxicity under normal use.

Chemical origin alone does not determine safety. Risk depends on the substance's toxicity, concentration, route of exposure, and duration of exposure.

Pesticides and Herbicides

Many pesticides and herbicides are synthetic organic chemicals designed to control:

- Weeds
- Insects
- Fungi
- Other unwanted organisms

Atrazine: A widely used triazine herbicide linked to endocrine disruption; regulated with a standard limit (Maximum Contaminant Level or MCL) of 3 µg/L (0.003 mg/L).

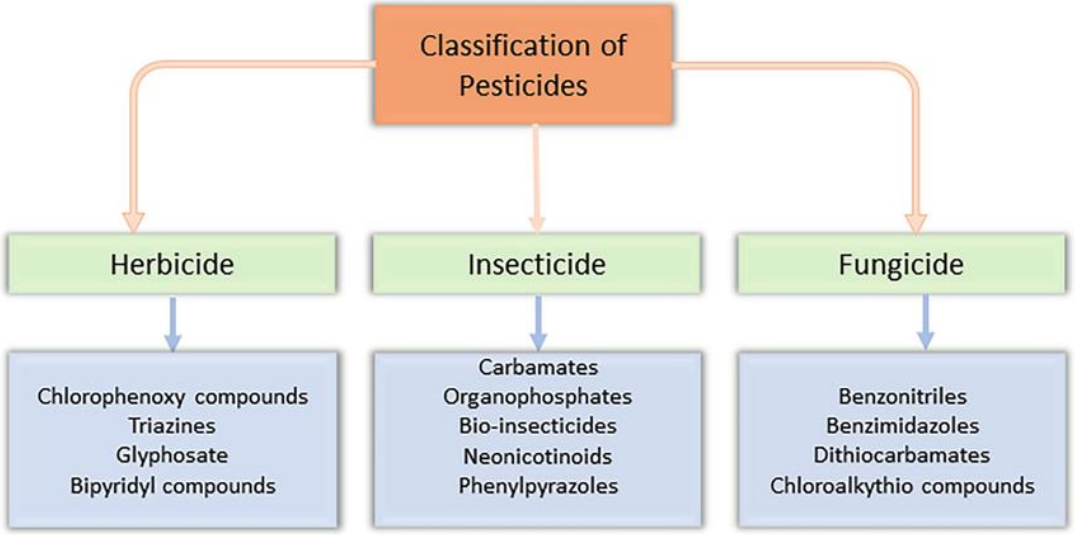
Alachlor: A chlorinatedacetamide herbicide monitored in agricultural runoff; regulated with an MCL of 2 µg/L (0.002 mg/L).

Simazine: A triazine compound frequently tracked in surface and groundwater testing near corn and-weed control sites.

2,4-D: A common chlorophenoxy systemic herbicide monitored for chronic toxic effe endocrine system

Pesticides can reach surface water or groundwater through:

- Agricultural runoff
- Leaching through soil
- Spills
- Improper storage
- Improper disposal



Organic Chemicals in the Environment

Stockholm Convention

Added 8 more substances to the agreement in 2009 and one more in 2011

New POPs Added to the Stockholm Convention in 2009 and 2011

POP	Use
Hexachlorocyclohexane	Unintentional by-product of lindane production
Chlordecone	Insecticide (ant and roach)
Hexabromobiphenyl	Fire-suppressing chemical
Hexabromodiphenyl and Heptabromodiphenyl ether	Flame retardant
Lindane	Broad-spectrum insecticide
Pentachlorobenzene	PCB
Perfluorooctane sulfonic acid	Key ingredient in Scotchgard fabric protector
Tetrabromodiphenyl ether	Flame retardant
Endosulfan	Crop insecticide

VOCs and Industrial Solvents

Volatile organic compounds, or VOCs, are organic chemicals that evaporate relatively easily.

EPA notes that many VOCs are human-made chemicals used in products and industrial processes.

Examples include:

- Trichloroethylene, or TCE
- Tetrachloroethylene, or PCE
- Benzene
- Some fuel-related compounds

Trichloroethylene — TCE

TCE has historically been used as an industrial solvent and metal degreaser.

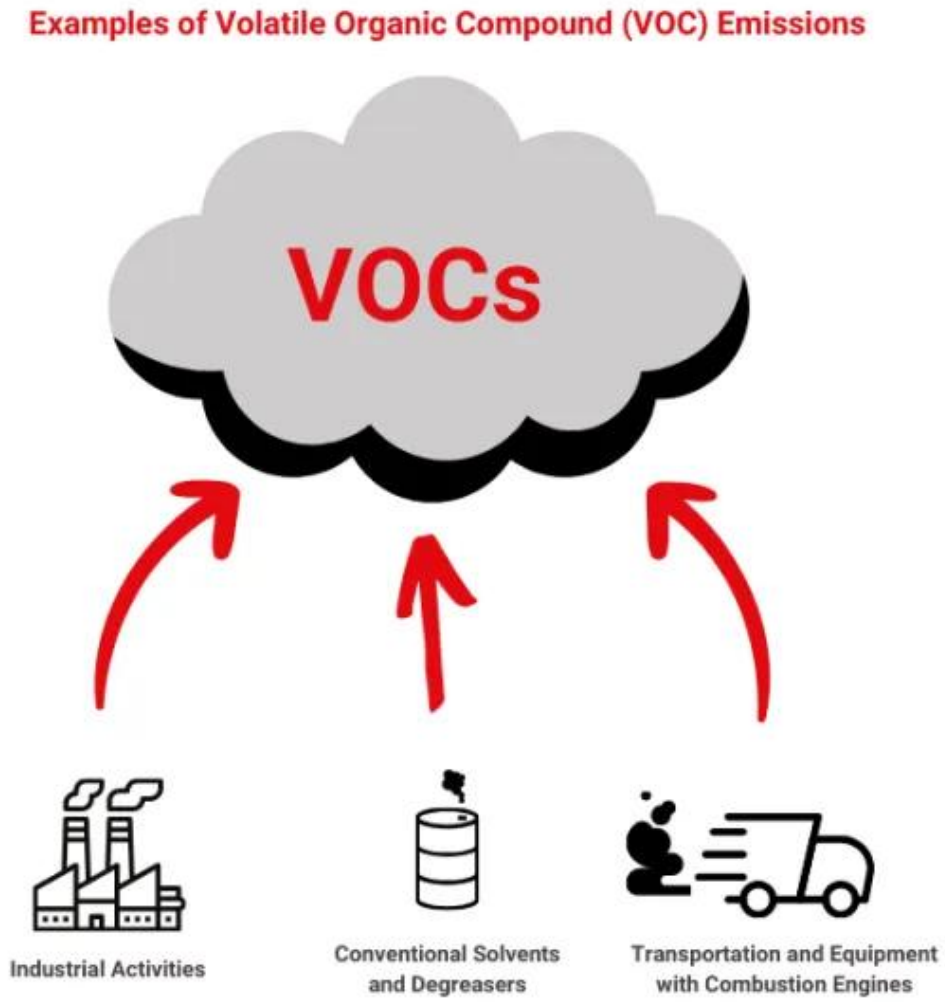
It can enter soil and groundwater through historical industrial releases, spills, and waste disposal.

EPA has concluded that TCE presents unreasonable risks under certain conditions of use, and exposure has been associated with multiple health concerns.

Tetrachloroethylene — PCE

PCE, also known as perchloroethylene, has been widely associated with dry cleaning and industrial degreasing.

EPA classifies PCE as **likely to be carcinogenic to humans**.



PFAS: Persistent Synthetic Organic Chemicals

PFAS are a large family of manufactured fluorinated chemicals.

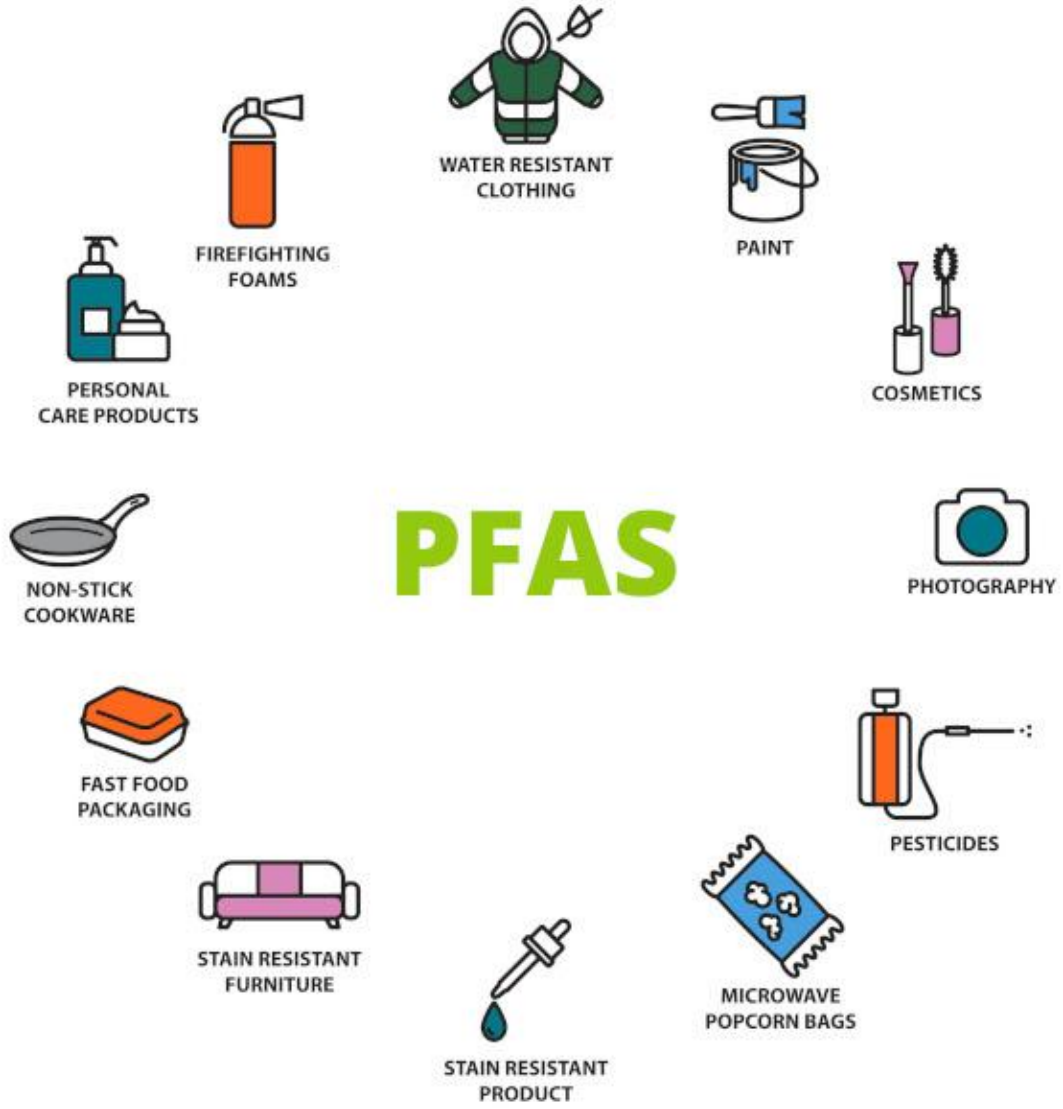
They have been used because some PFAS provide resistance to:

- Water
- Oil
- Grease
- Heat

PFAS have been associated with applications such as

- Firefighting foams
- Industrial processing
- Stain-resistant materials
- Certain food-contact materials
- Water-resistant products

Many PFAS are extremely persistent.



How Do Synthetic Organic Chemicals Reach Drinking Water?
SOCs can reach water through a variety of pathways.

Agricultural Runoff
Pesticides applied to agricultural land can move into rivers or groundwater through runoff and leaching.

Industrial Releases
Historical or current industrial activities can release solvents and other chemicals into:
Soil
Groundwater
Surface water

Landfills
Water moving through discarded materials can create leachate containing a complex mixture of dissolved contaminants.

Wastewater
Household and industrial chemicals can enter wastewater collection systems.
Treatment effectiveness depends on the chemical and process.

Atmospheric Deposition
Some chemicals can travel through the atmosphere before eventually depositing onto land or water.

Clear water can still contain trace organic contaminants. Most SOCs relevant to drinking-water concerns cannot be identified reliably by sight, smell, or taste.

How Can SOCs Affect Human Health?

There is no single health profile for synthetic organic chemicals.
Potential effects depend on the individual chemical.

Endocrine Effects

Some chemicals can interfere with hormone signaling.
Examples studied for possible endocrine effects include:

- Some phthalates
- BPA
- Some pesticides
- Some PFAS

Cancer

Some synthetic organic chemicals are known or suspected carcinogens.
Others are not.
For example, specific industrial solvents such as TCE and PCE have significant cancer-related toxicological evidence.

Liver, Kidney, and Neurological Effects

Some solvents, pesticides, and other compounds can affect organs or the nervous system at sufficient exposures.

Developmental or Reproductive Effects

Certain chemicals have been studied for effects on:

- Fetal development
- Fertility
- Hormone regulation
- Child development

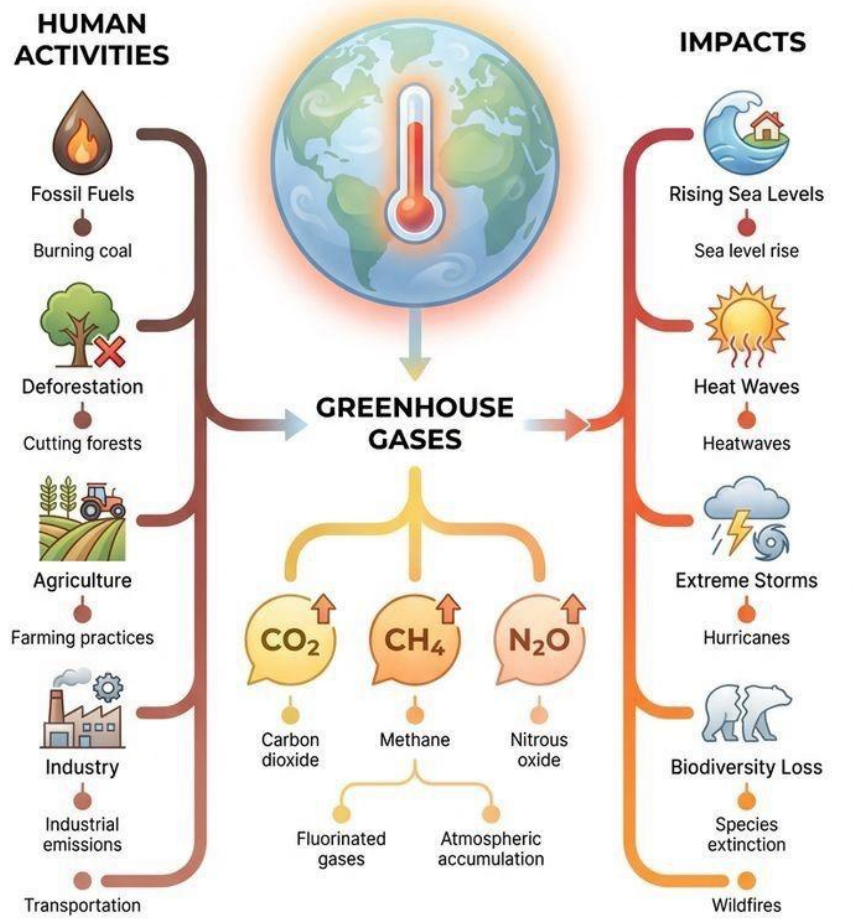
Organic Chemicals in the Environment

Stockholm Convention

Added 8 more substances to the agreement in 2009 and one more in 2011

- Five of the new POPs are also poly chlorinated compounds which are used as insecticides
- Polychlorinated hydrocarbons are insoluble in water but soluble in fats
- Polychlorinated hydrocarbons concentrated in the fatty tissues of fish and of birds, and of humans who eat them
- For the first time, polybrominated compounds, used in the 1970s as flame-retardant agents, were added to the list.
- Perfluorooctane sulfonic acid, the key ingredient in Scotchgard fabric protector, was added to the list of new POPs

CLIMATE CHANGE: CAUSES & EFFECTS



POSSIBLE SOLUTIONS

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Organic Chemicals in the Environment

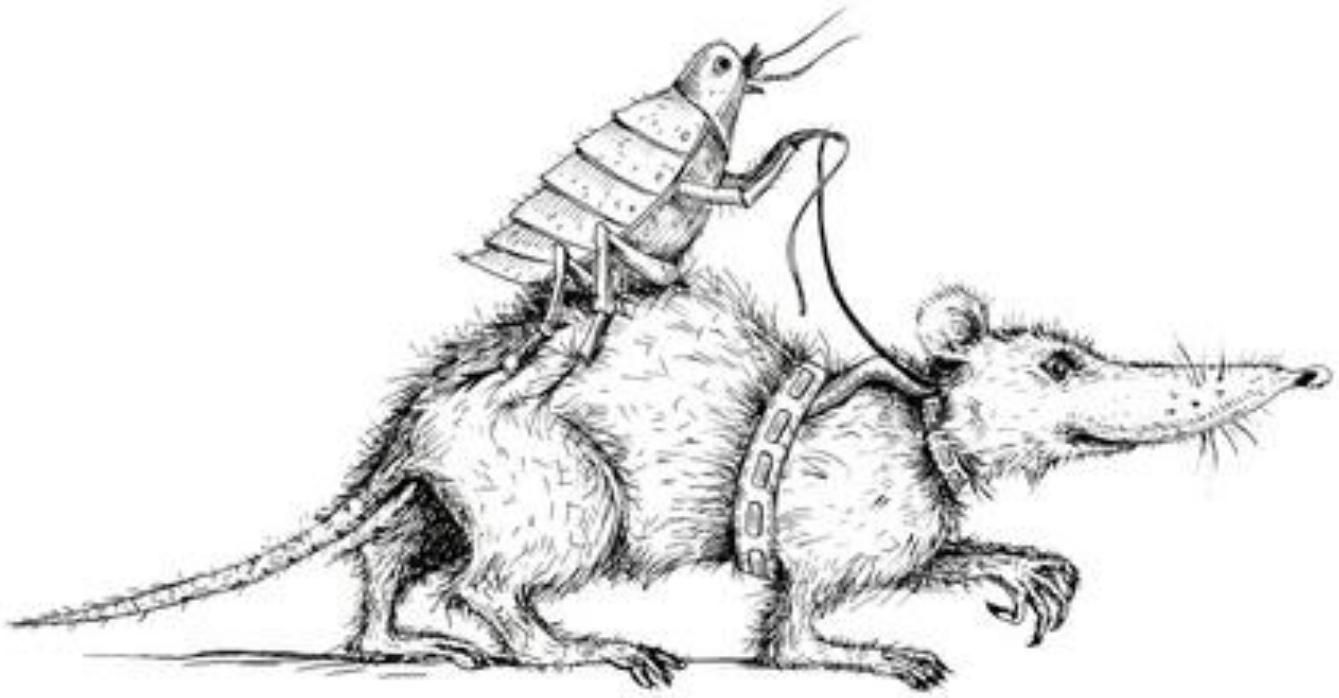
Organic Chemicals in the Environment

Why are we producing dangerous synthetic chemicals intentionally?

- Human civilization always have had to contend with pests

- **The three most impactful pests in human history:**

1) Rats and fleas, which spread the devastating Black Death (1347-1350)



Black Death: an infectious fever caused by the **Bacterium Yersinia pestis**. The disease was likely transmitted from **rodents** to humans by the bite of infected **fleas**

Organic Chemicals in the Environment

Why are we producing these dangerous chemicals intentionally?

- Human civilization always have had to contend with pests

- **The three most impactful pests in human history:**

1) Rats and fleas, which spread the devastating Black Death (1347-1350);
2) Mosquitoes, which carry diseases like malaria and yellow fever;
3) Viruses, responsible for smallpox and Spanish flu

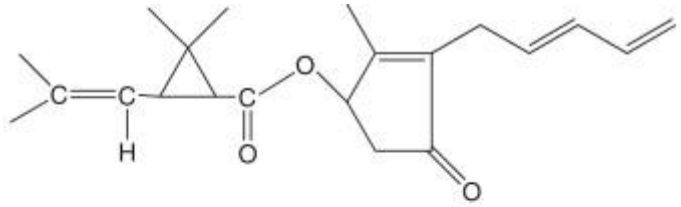
Organic Chemicals in the Environment

What were we using before Synthetic Pesticides came into use?

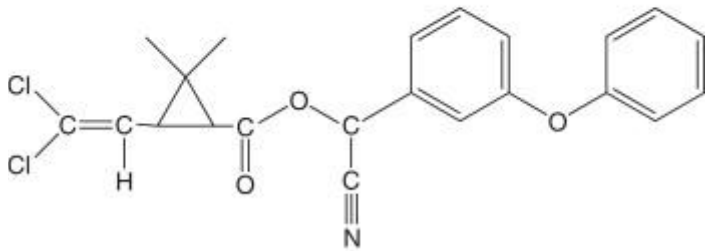
- Before 1940, only a few pesticides were available and most of the insect poisons naturally extracted from plants

Examples:

1. Pyrethrins obtained from the pyrethrum flower



Pyrethrin I



Cypermethrin

Organic Chemicals in the Environment

Companion Plantation of pyrethrum plants in **organic farming**

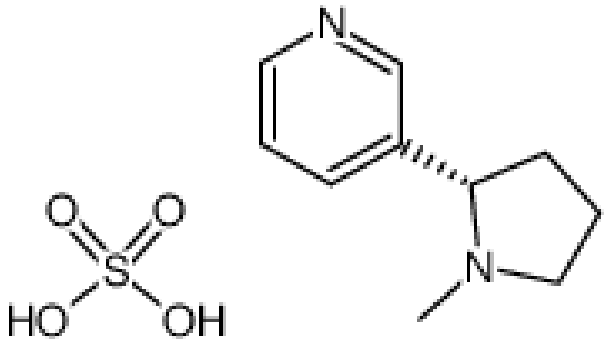
Organic Chemicals in the Environment

What were we using before Synthetic Pesticides came into use?

- Before 1940, only a few pesticides were available and most of the insect poisons naturally extracted from plants

Examples:

2. Nicotine sulfate; Nicotine obtained from tobacco plants is neutralized with sulfuric acid



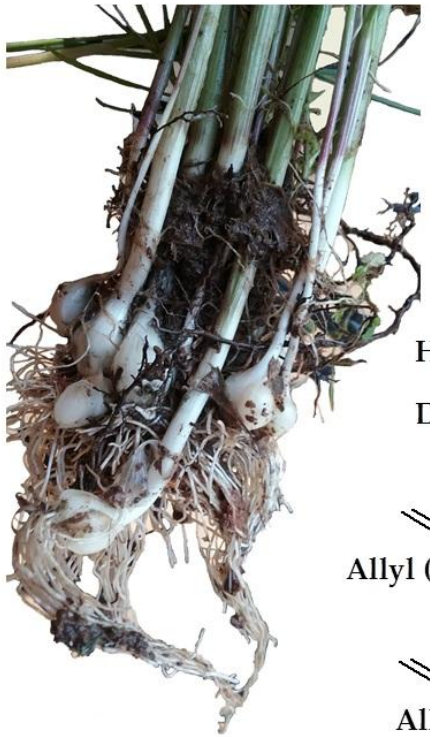
Organic Chemicals in the Environment

What were we using before Synthetic Pesticides came into use?

- Before 1940, only a few pesticides were available and most of the insect poisons naturally extracted from plants

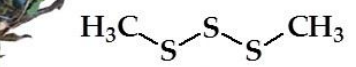
Examples:

3) Garlic oil

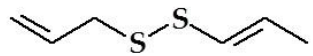


Allium vineale

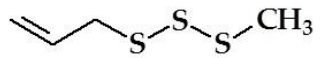
Chemical Compositions of the Volatile Oils of Garlic



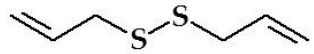
Dimethyl trisulfide



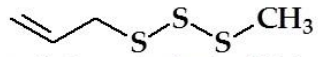
Allyl (*E*)-1-propenyl disulfide



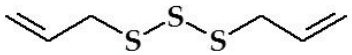
Allyl methyl trisulfide



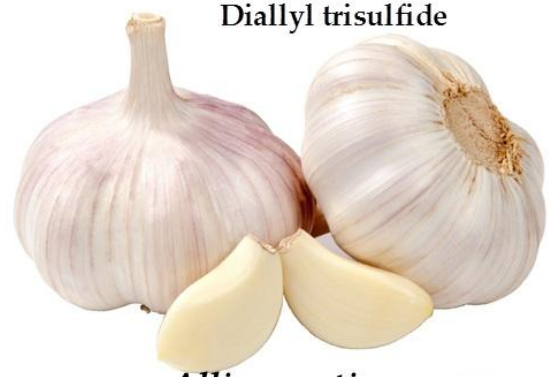
Diallyl disulfide



Allyl methyl trisulfide



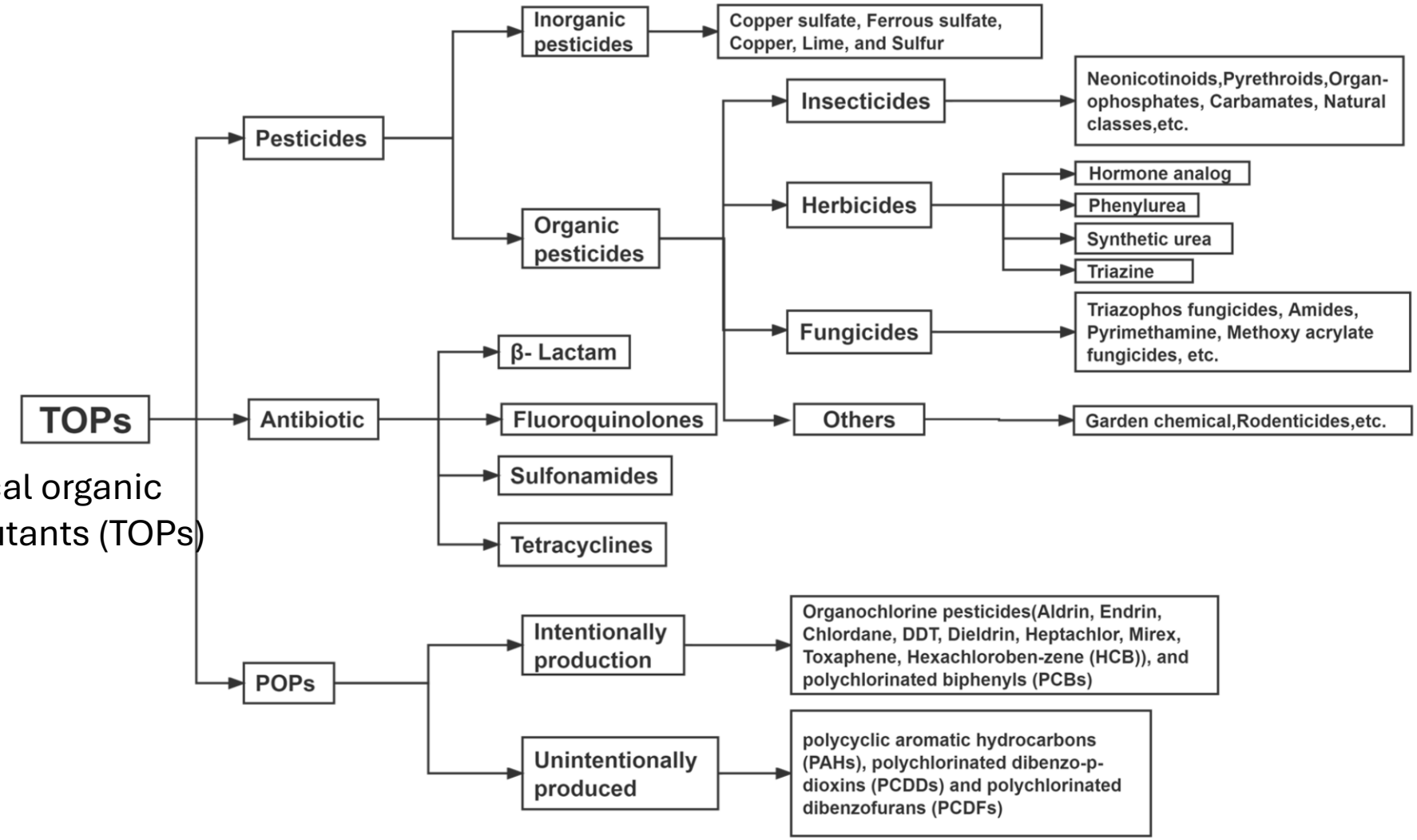
Diallyl trisulfide



Allium sativum

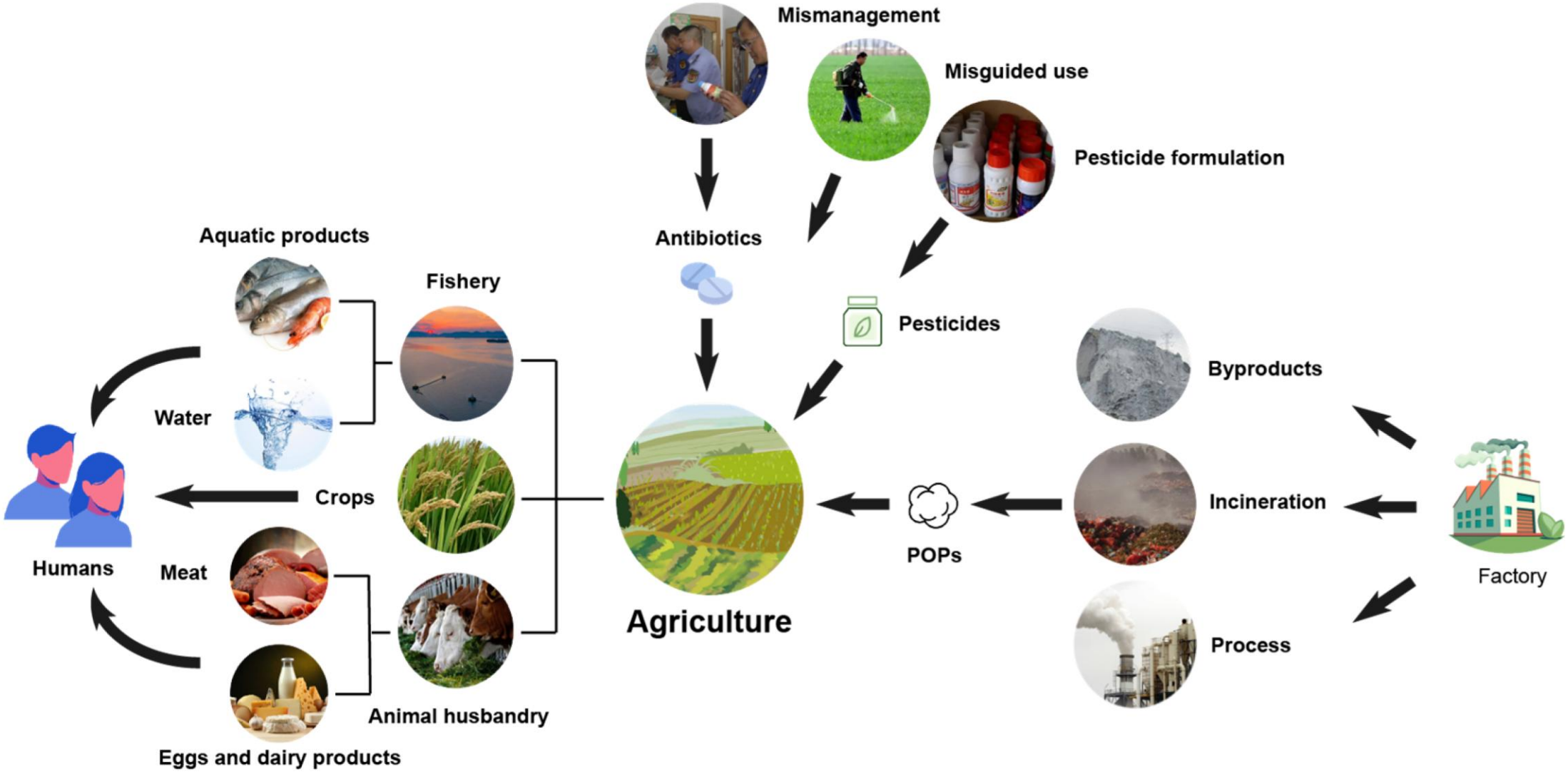
The classification of typical organic pollutants in the agricultural environment

typical organic pollutants (TOPs)



https://doi.org/10.3390/agronomy12123009

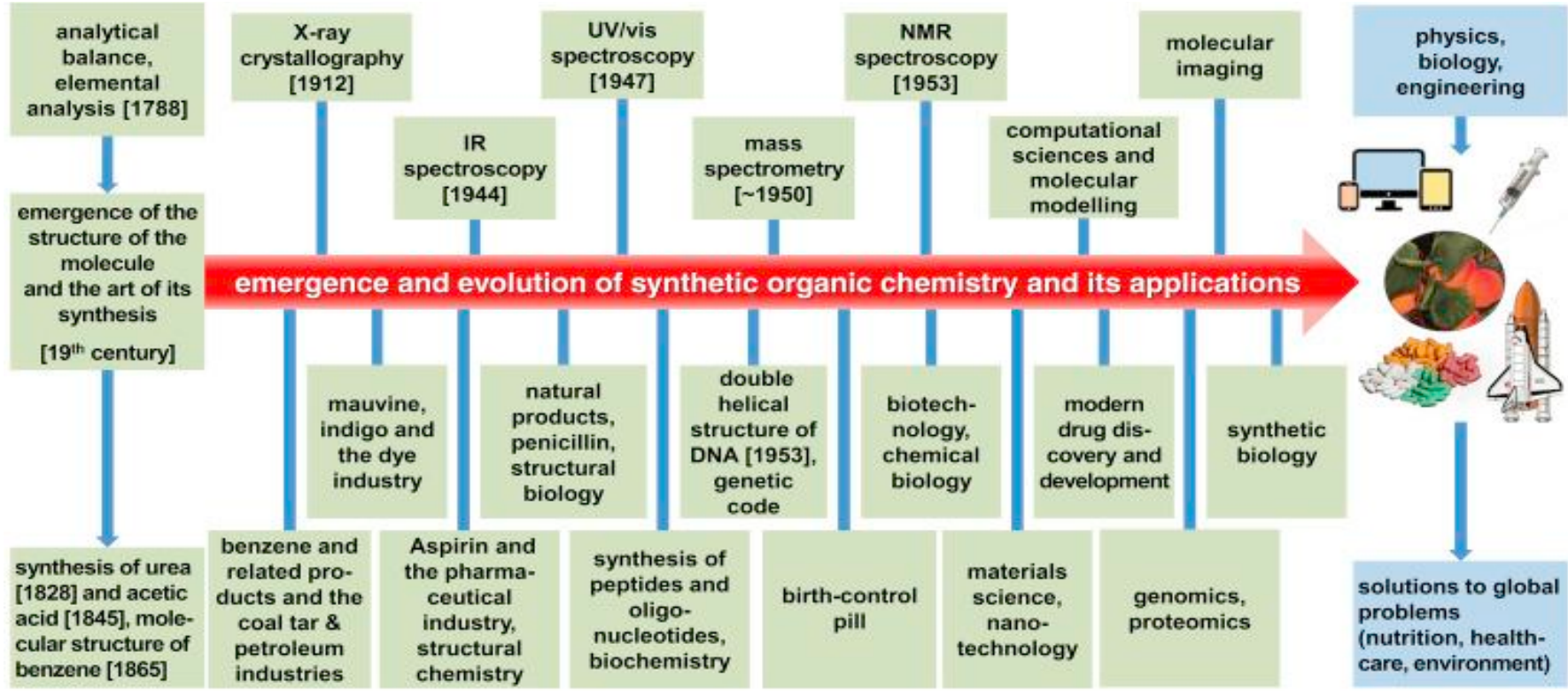
The sources and spread of typical organic pollutants in the agricultural environment.



https://doi.org/10.3390/agronomy12123009

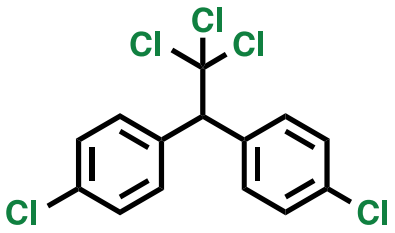
Synthetic Organic Chemicals in the Environment

Synthetic organic chemicals (SOCs) are man-made, carbon-based compounds derived from petroleum, natural gas, or coal that do not occur naturally in the environment.



Chem 2016 1331-334DOI: (10.1016/j.chempr.2016.08.006)

Synthetic Organic Chemicals in the Environment



DDT
(Dichlorodiphenyltrichloroethane)

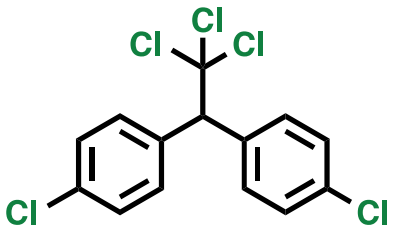
- The introduction of DDT at the end of WW-II (late 1940s) escalated the use of pesticides and the development of several other pesticides

Advantages

- Inexpensive
- Nontoxic to humans and other mammals
- Stable
- Continued to kill insects for a long period after application

Discovery and History
Created in 1874: Austrian chemist Othmar Zeidler first made the chemical.
Found useful in 1939: Swiss chemist Paul Hermann Müller discovered that it killed insects. He won a Nobel Prize for this in 1948.

Synthetic Organic Chemicals in the Environment

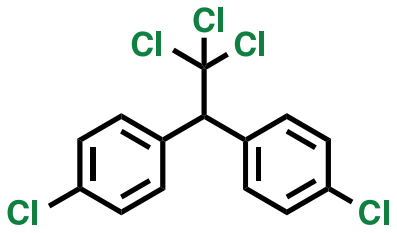


DDT
(Dichlorodiphenyltrichloroethane)

Disadvantages

- Harmful to fish-eating birds such as bald eagles, peregrine falcons, and brown pelicans
- Water gets contaminated with DDT from aerial spraying and runoff from treated land
- With time, insects also became increasingly resistant to DDT and larger amounts had to be applied to achieve the desired results
- DDT interferes with calcium metabolism in birds and it causes eggshells to become thin and break when parent birds attempt to incubate the eggs

Synthetic Organic Chemicals in the Environment

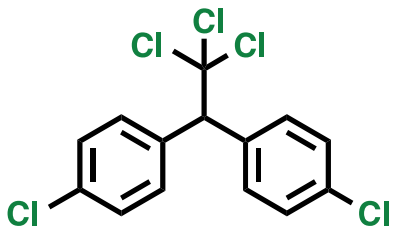


DDT
(Dichlorodiphenyltrichloroethane)

How does DDT get into food chain if it is insoluble in water?

- Contaminated runoff seeps into water bodies, then since the DDT is insoluble in water, it settles in the sediments.
- Plankton absorb it from the sediments in water.
- This DDT is then transferred to small fish when they eat the plankton, and the concentration increases further as larger fish consume the small fish, leading to biomagnification.
- The highest concentrations of DDT accumulate in the top predators, such as fish-eating birds or humans.

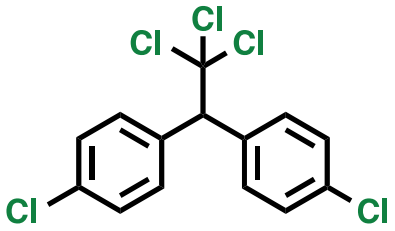
Synthetic Organic Chemicals in the Environment



DDT
(Dichlorodiphenyltrichloroethane)

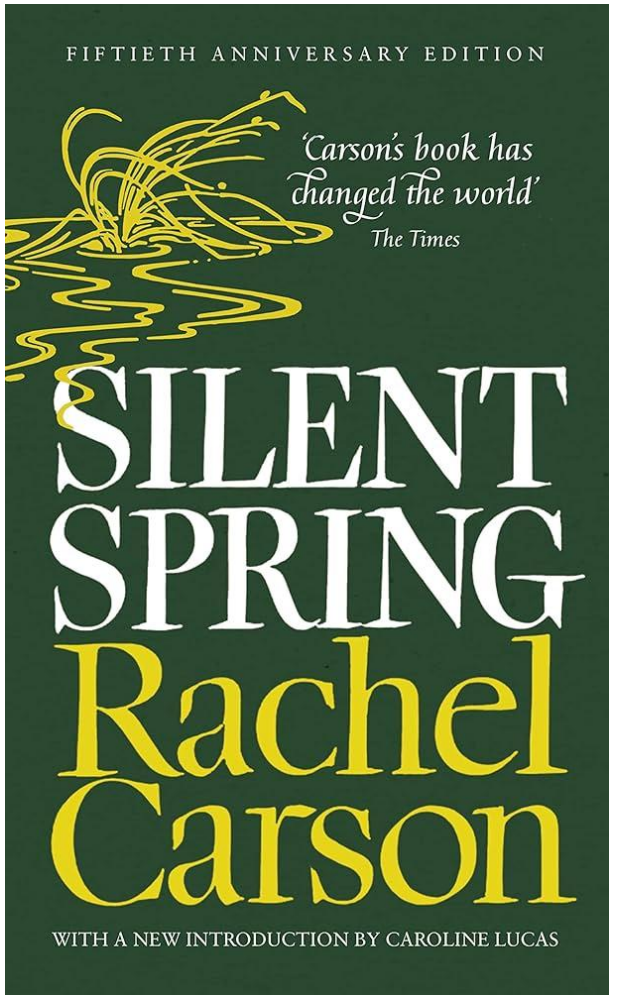
- The first person to draw public attention to the dangers of pesticides was biologist **Rachel Carson** and she played a key role in the establishment of Environmental Protection Agency (EPA) in 1970
- In 1973, the use of DDT was banned in the United States and fish-eating birds population restored shortly
- Still DDT is partially banned in several countries such as India
- DDT is banned for agriculture but allowed for fumigation to kill mosquitoes to prevent malaria

Synthetic Organic Chemicals in the Environment



DDT

(Dichlorodiphenyltrichloroethane)



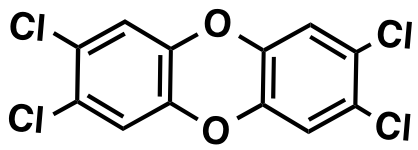
Environmental Scientist

Vs

Environmental Activist

Synthetic Organic Chemicals in the Environment

Dioxins 2,3,7,8-tetrachlorodibenzo-*p*-dioxin (TCDD)

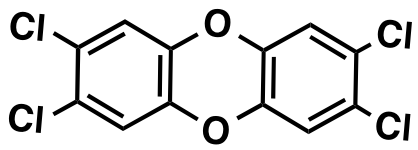


Produced by burning chlorine containing medical and municipal wastes

- TCDD persists in the environment for a very long time and bioaccumulates within the food chain
- TCDD is extraordinarily toxic to guinea pigs and is a known carcinogen for many animals
- During Vietnam war, USA used **Agent Orange** (defoliant) in which TCDD present as an impurity
- EPA concluded that exposure to dioxin, increase the risk of cancer, may disrupt some reproductive mechanisms and suppress the immune system.

Synthetic Organic Chemicals in the Environment

Dioxins 2,3,7,8-tetrachlorodibenzo-*p*-dioxin (TCDD)



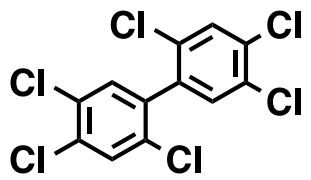
Major sources for TCDD

- Burning of chlorine-containing **medical and municipal** wastes
- **Paper mills** that use chlorine to bleach paper pulp
- During the manufacture of **trichlorophenol** a precursor for herbicides

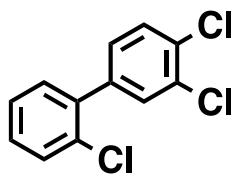
Currently, proper solutions were found for above three problems

Synthetic Organic Chemicals in the Environment

PCBs (Polychlorobiphenyls)

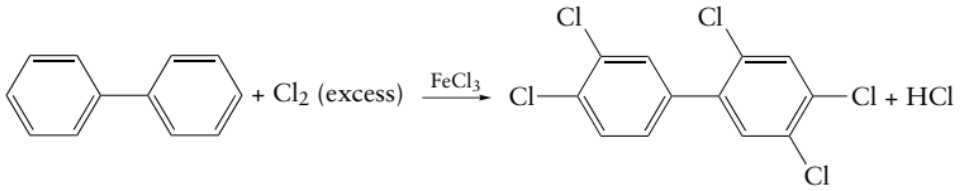


PCB_1



PCB_2

- PCBs are not water soluble but readily soluble in fat
- PCBs are chemically and thermally stable, fire and electricity resistant and persist in the environment for many years



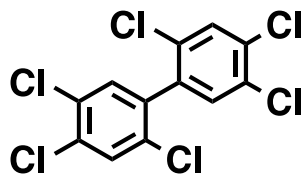
(One of many products)

Uses of PCBs

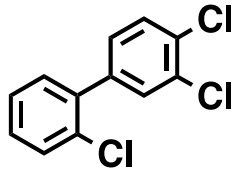
- Heat exchange fluid in transformers,
- Paints and plastic additive (plasticizer)

Synthetic Organic Chemicals in the Environment

PCBs (Polychlorobiphenyls)



PCB_1



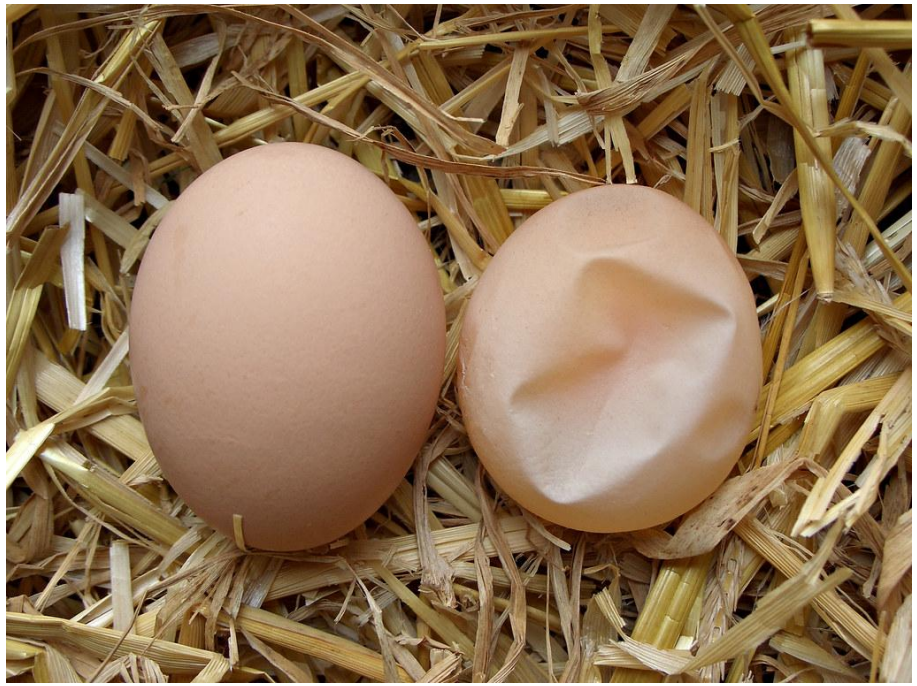
PCB_2

- When plastic wastes and other PCB-containing materials are burned, PCB vapors condense on airborne particles, which then fall directly onto water or reach water in runoff from the land

- PCBs present in water in a negligible concentration but can become more than 1 million times concentrated in fish

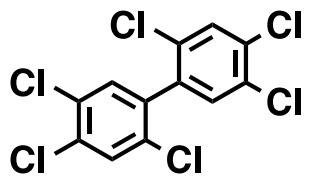
- PCBs cause eggshell thinning and neurologic damage in birds and impair reproduction of aquatic species

- In humans, they cause, chloracne, liver damage, reproductive disorders, birth defects, and cancer

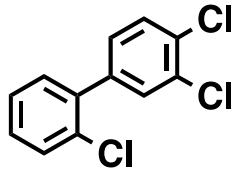


Synthetic Organic Chemicals in the Environment

PCBs (Polychlorobiphenyls)



PCB_1



PCB_2

- In 1977, production of PCBs in the United States was halted, and disposal of PCB-containing products is now strictly regulated
- Fish in many lakes still contain significant levels of contaminants
- Unfortunately, PCBs are still produced in Russia and many developed countries

Goals of PBT program - 4

PBT stands for **persistent, bioaccumulative, and toxic** chemicals. These substances do not readily break down in the environment, can accumulate in living organisms and food chains, and may cause harmful effects in humans, wildlife, and ecosystems. The U.S. Environmental Protection Agency's PBT program is designed to manage and reduce these long-term risks.

The four goals shown are:

Prevent new PBT chemicals from entering the market: Evaluate and regulate potentially hazardous chemicals before they are widely produced or used.

Reduce exposure to existing priority PBTs: Minimize human and environmental contact with PBT chemicals that are already present in air, water, soil, consumer products, or waste streams.

Stop their movement through the environment: Limit transport of PBTs through air, water, and land, since they can travel long distances and contaminate remote regions.

Understand long-term environmental effects: Study how these chemicals persist, accumulate, spread, and affect ecosystems over many years or decades.

Study of Environmental fate of Organic Chemicals

- Water solubility
- Soil/sediment sorption coefficient
- Bioconcentration

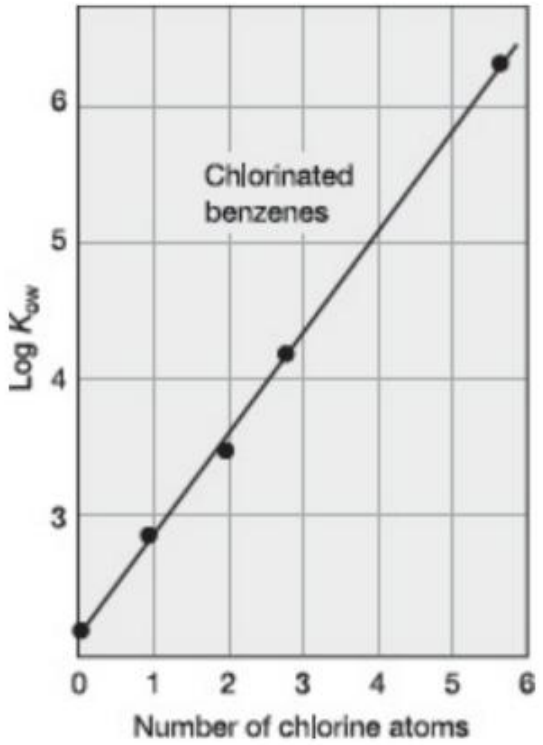
These properties help predict whether a chemical will remain dissolved in water, stick to soil or sediment, or enter and accumulate in organisms

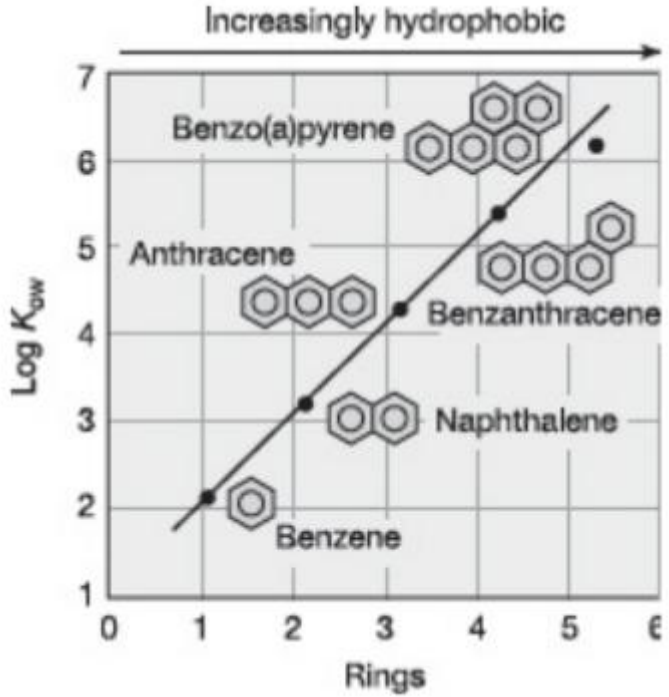
Water solubility is most important – it is the governing factor to decide on the fate and transport of chemicals. If the compound is readily soluble , hydrolysis and photolysis events occur spontaneously

Octanol/water partition coefficient (K_{ow}) : ratio of molar concentrations of a chemical in n-octanol and water

K_{ow} ratio expresses the tendency of an organic contaminant to move from water phase to immiscible 1-Octanol phas
Long chain alcohol mimics the behavior of fats (lipids)
K_{ow} is an indication of the tendency of an organic contaminant to bioaccumulate

Environmental Assessment based on K_{ow}





Which one has a more tendency to degrade?

Compound	Log K_{ow}
aldrin	6.5
DDT	6.2
hexachlorobenzene	5.6
chlordane	5.5
heptachlor	5.3
toxaphene	4.8
fluoranthene	4.7
Endrin	4.6
phenanthrene	4.5
Dieldrin	4.3
1,2,4-trichlorobenzene	4.2
dibutyl phthalate	4.1
dibenzyl	4.0
1,4-dichlorobenzene	3.6
ortho-dichlorobenzene	3.7
chlorobenzene	2.8
toluene	2.7
benzene	2.1
phenol	1.5

PBT profile is intended to answer three questions for manufacturers?

1. Once released, will a chemical go into the air, water, soil, or sediment?
2. How long will this chemical stay in the medium (air, water, soil or sediment)?
3. Will the chemical present hazard?

PBT profiler calculates an atmospheric half-life by determining the importance of a chemical's reaction with two of the most prevalent atmospheric oxidants, **hydroxyl radicals** and **ozone**.

The half-life is calculated directly from gas phase hydroxyl radical and ozone reaction rate constants.

The 12 EPA's Persistence Criteria for PBT Chemicals

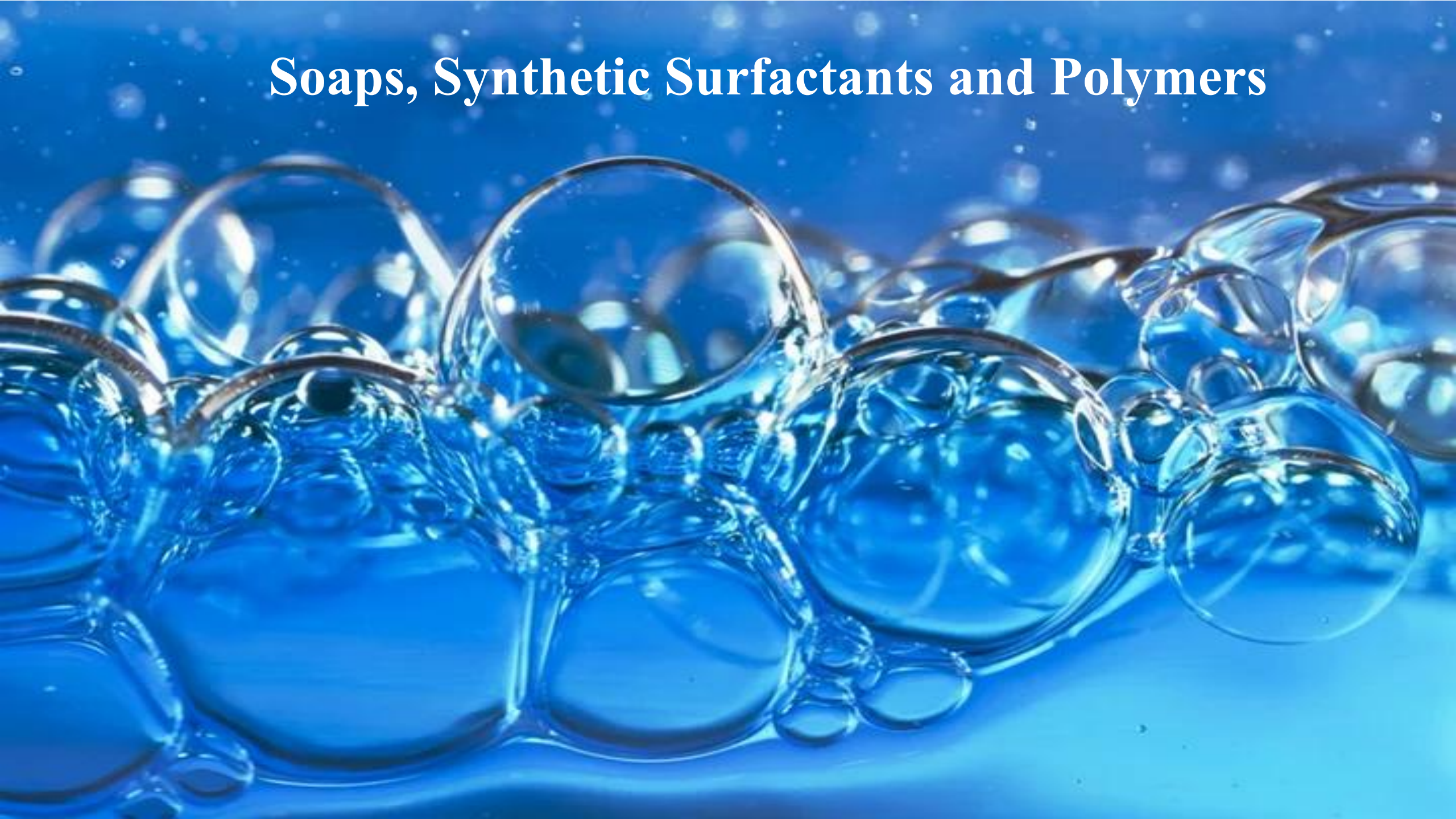
	Half-Life of Chemical		
Media	Not Persistent	Persistent	Highly Persistent
Water	< 2 months	≥ 2 months	> 6 months
Soil	< 2 months	≥ 2 months	> 6 months
Air	≤ 2 days		> 2 days
Sediment	< 2 months	≥ 2 months	> 6 months

Under **Toxic Substances Control Act (TSCA)**, chemical manufacturers are required by law to submit **premanufacture notifications (PMNs)** to the EPA for new chemicals that they are planning to manufacture domestically or import on an industrial scale.

The EPA uses a sophisticated computer program that evaluates a chemical's characteristics to ascertain whether it might pose an unreasonable risk to human health or the environment.

Based on this evaluation, the EPA can stop the production of these chemicals until the manufacturer can prove that they will not pose an unreasonable risk if released into the environment.

Soaps, Synthetic Surfactants and Polymers



Soaps, Synthetic Surfactants and Polymers

Most of the polymers and surfactants used in detergents are mainly made from chemicals derived from petroleum

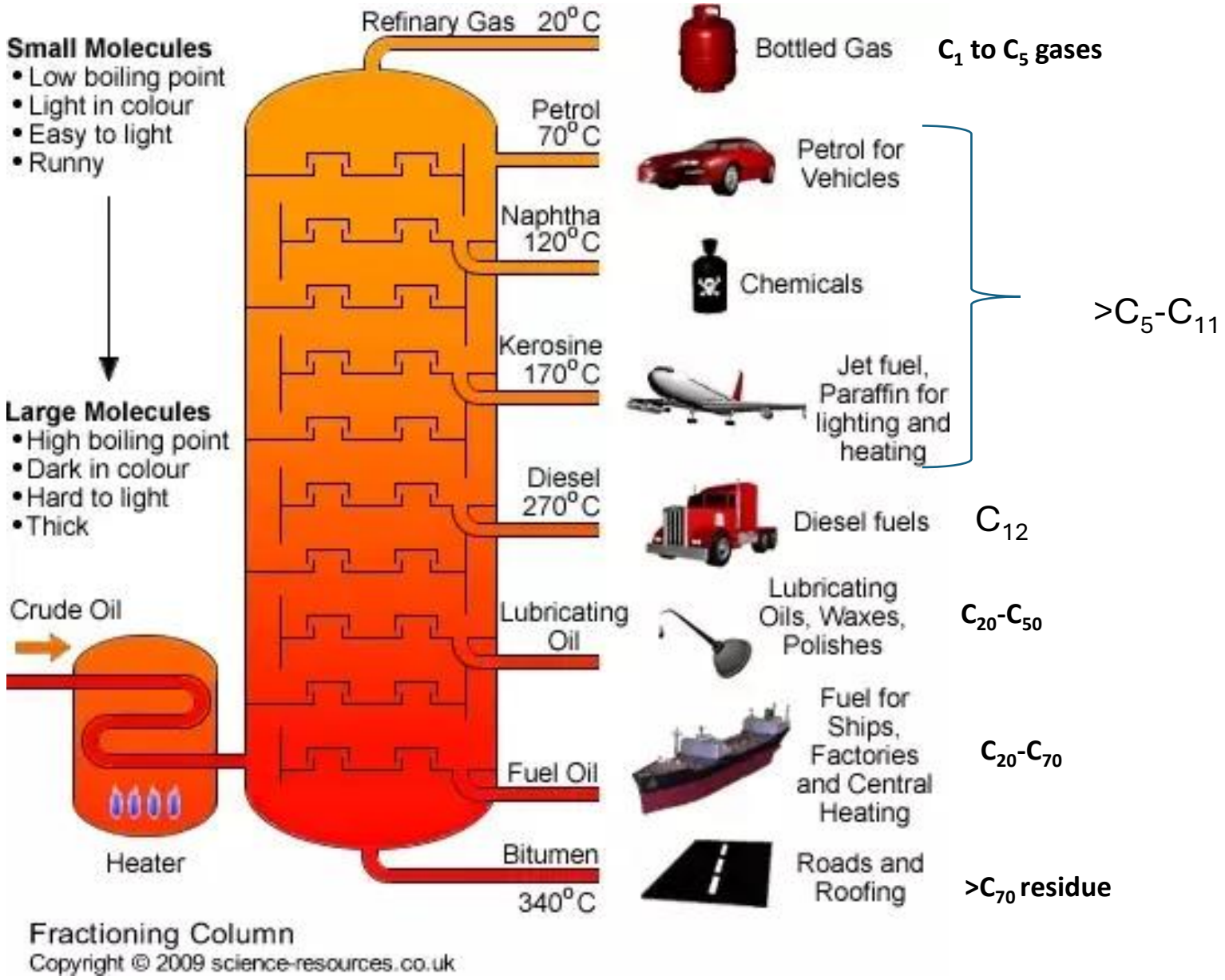
Many of carbon compounds that have been formed by geochemical reactions (crude oil, coal) or industrial chemical processes (polyethylene, DDT) are not readily degradable and are frequently referred to as "pollutants"

Environmental problem: The rate of degradation by environmental reagents such as sunlight, oxygen or water is much slower than that of microbial degradation

Fractional Distillation of Crude Oil

- Oil usually occurs with natural gas: mixture of hydrocarbons of light molecular weight

Crude petroleum is a complex mixture of hydrocarbons that is separated into fractions of differing boiling ranges by the refining process shown schematically in Figure 6-1. The various fractions either are used directly as energy sources ("straight-run gasoline") or are modified chemically to yield more efficient energy sources such as high-octane gasoline. In addition, hydrocarbons may be converted to petrochemical compounds such as monomers, polymers, and detergents that are derived from petroleum.

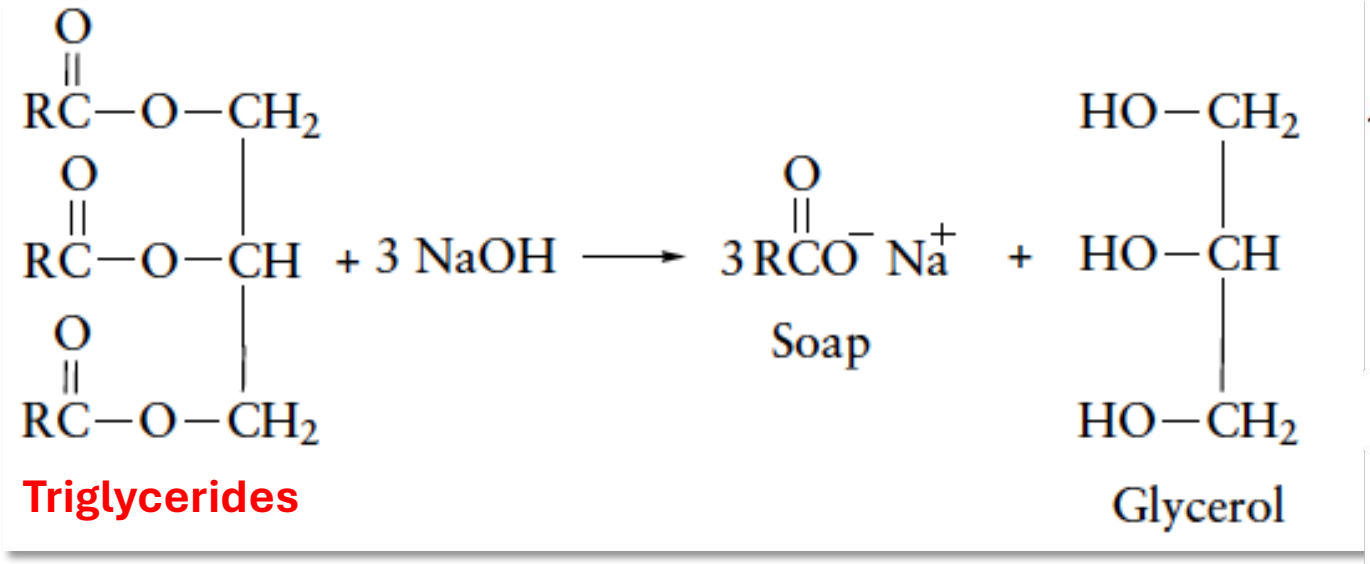


Organic Chemicals in the Environment

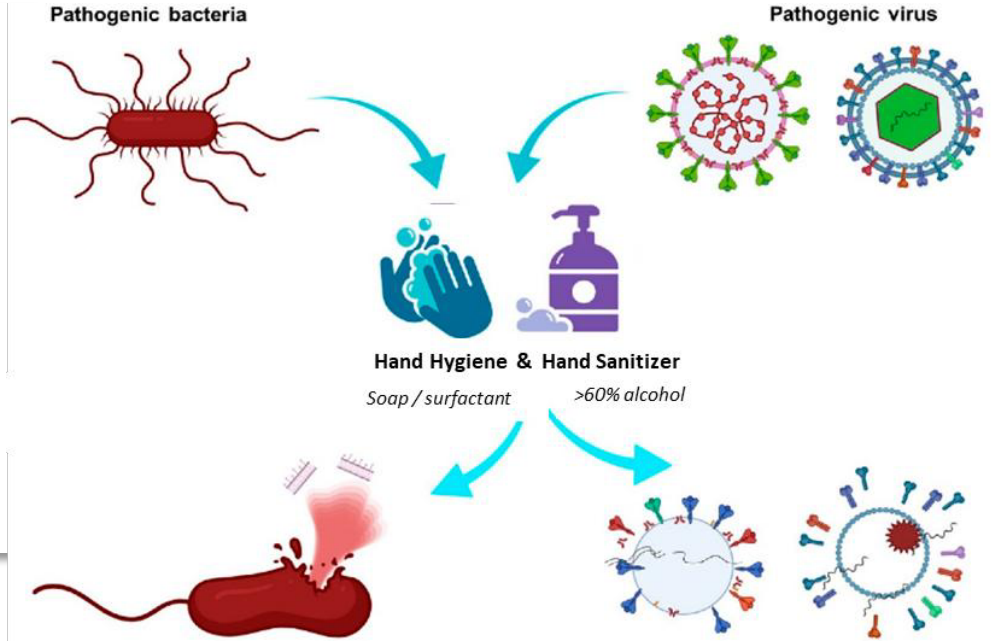
Soaps and Synthetic Surfactants

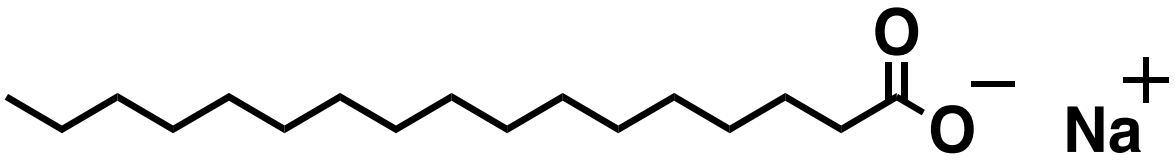
Soap: Sodium salts of fatty acids

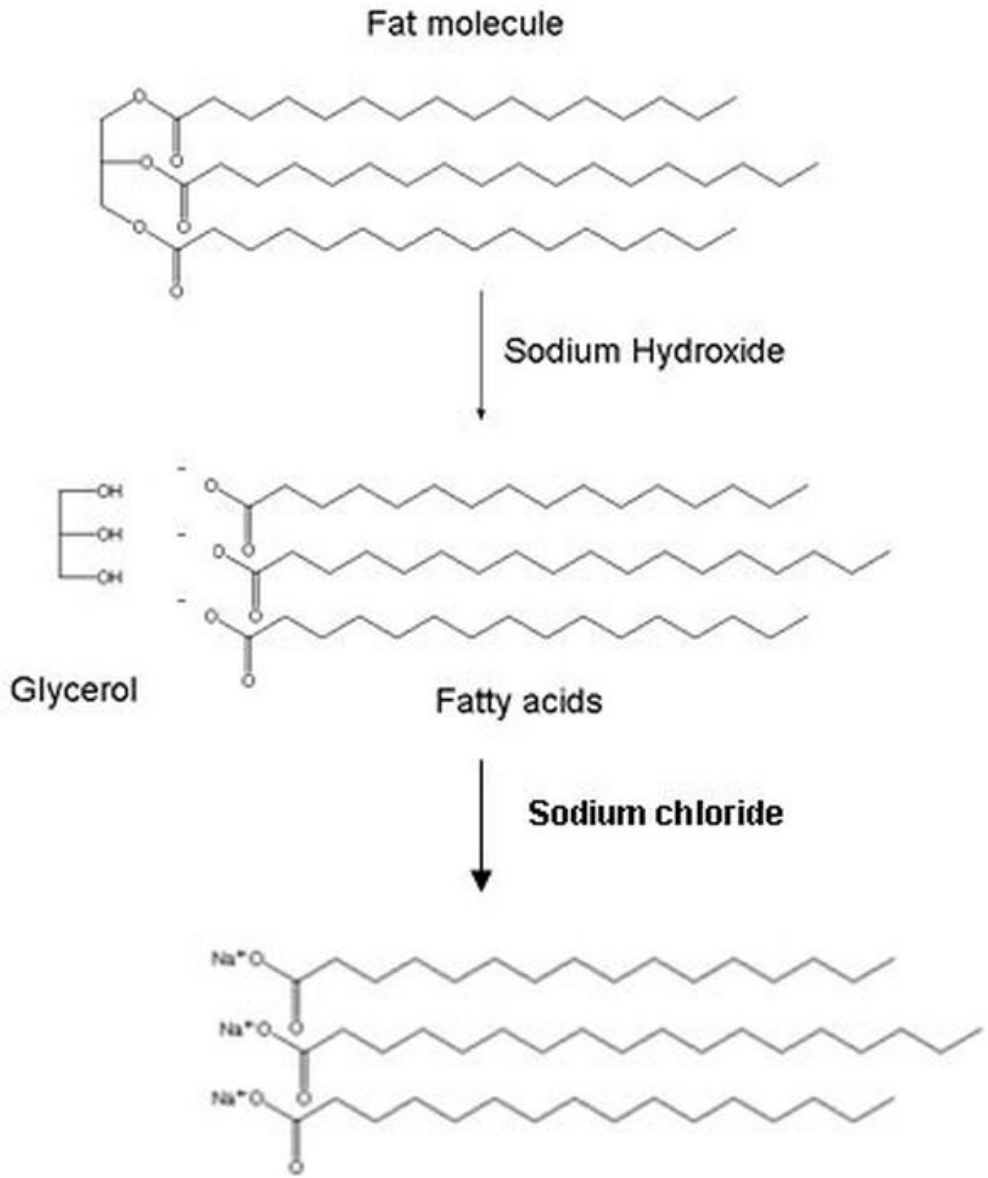
Prepared by the saponification (base hydrolysis) of animal fats or vegetable oils



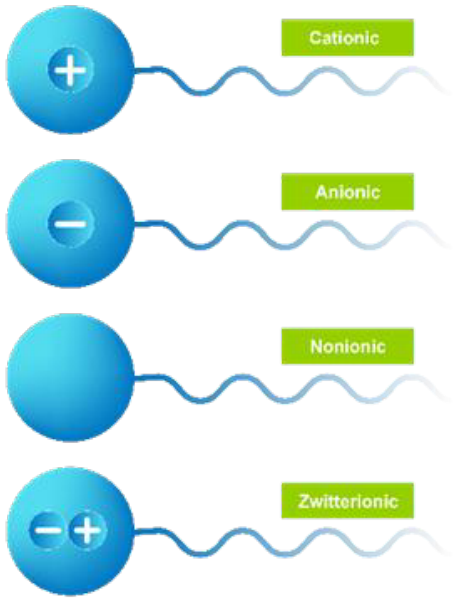
Triglycerides






Fat molecule
Sodium Hydroxide
Glycerol
Fatty acids
Sodium chloride
Fatty acid salts (soap)

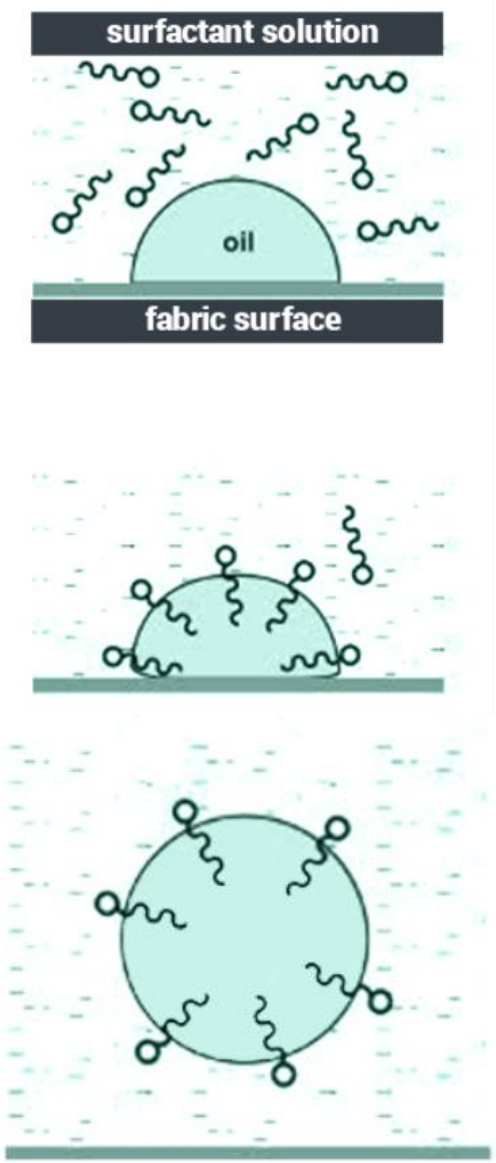
Hydrophilic (water-loving) head
Hydrophobic (water-hating) tail


Cationic
Anionic
Nonionic
Zwitterionic

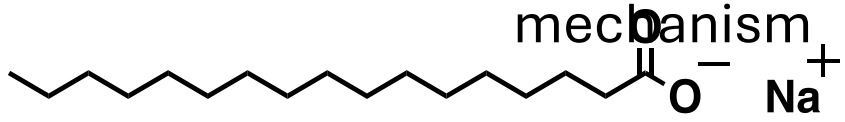
Surfactant Containing Products
Hand Wash
Dish Washing Soap
Liquid Soap
Liquid Detergent

Organic Chemicals in the Environment

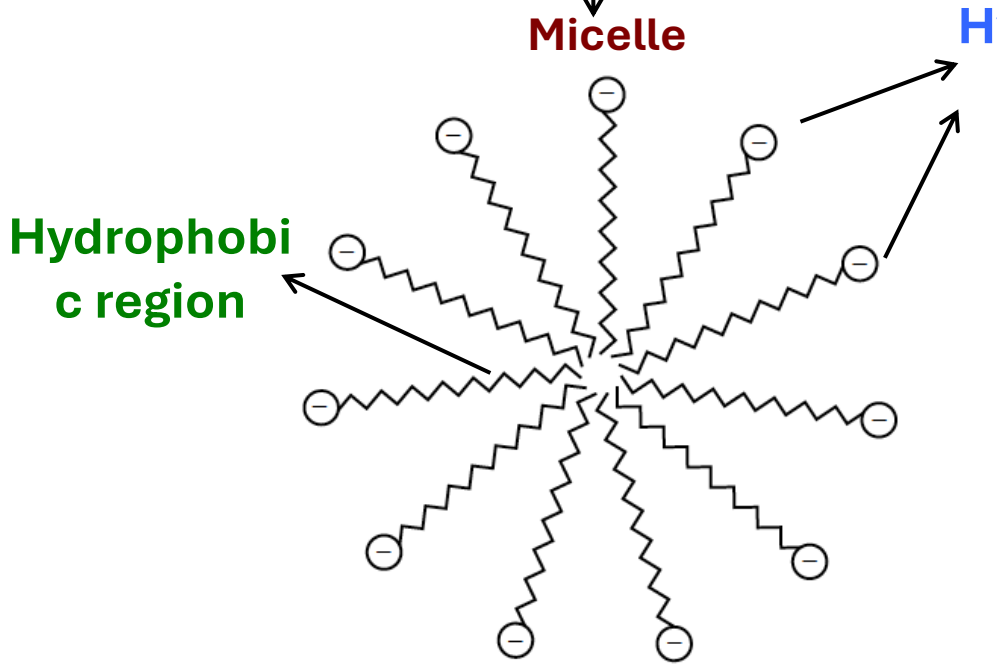
Soaps and Synthetic Surfactants



Mechanism of cleansing: Both soaps and surfactants operate by similar cleansing mechanism



H_2O



Hydrophobic portion is attached with hydrophilic group. Such materials are surface active. i.e., they concentrate at the surface of an aqueous solution or form aggregates called micelles within solution

Organic Chemicals in the Environment

Soaps and Synthetic Surfactants

What are soaps and surfactants? Contain segments of linear or lightly branched hydrocarbon Chains, that are broken down to acetate by metabolism by microorganisms in the environment

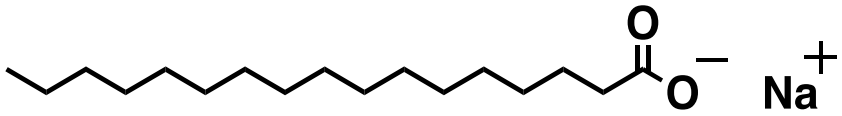
- The sodium salt of stearic acid, $CH_3(CH_2)_{16}CO_2H$, is the major product released upon hydrolysis of animal fat

- The sodium salt of oleic acid, $CH_3(CH_2)_7CH{=}CH(CH_2)_7CO_2H$, is the major product released upon hydrolysis of olive oil

- Hydrolysis of palm oil yields approximately equal amounts of the salts of oleic acid and palmitic acid, $CH_3(CH_2)_{14}CO_2H$.

Organic Chemicals in the Environment

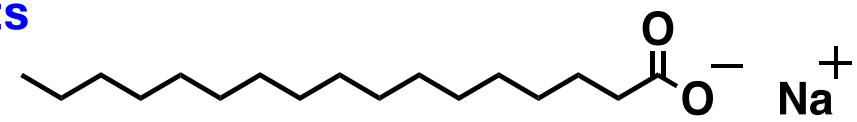
Soaps and Synthetic Surfactants



- All these are naturally occurring fatty acids and contain linear hydrocarbon chains.
- They are synthesized biochemically from the acetate ion ($CH_3CO_2^-$).
- They are also readily degradable back to acetate by microorganisms in the environment.

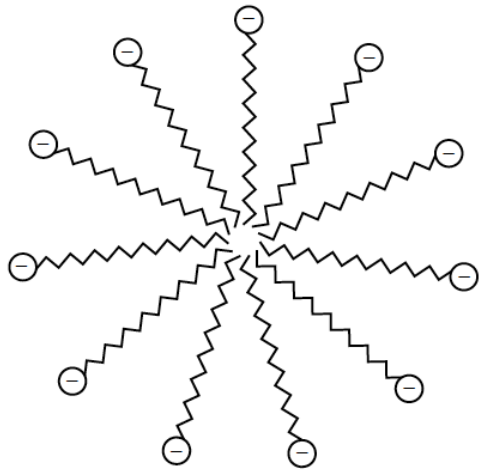
Organic Chemicals in the Environment

Soaps and Synthetic Surfactants



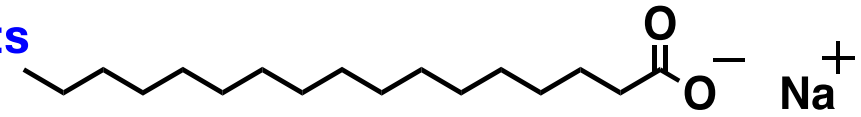
Disadvantages

- Carboxylic acid salts form insoluble precipitates with Ca^{2+} and Mg^{2+} in water (hard water)
- Weak acidic nature of carboxylic acids allows protonation even in mildly acidic solutions
- Polarity of the head group decreases when protonated and solubilizing ability of oils and greases is diminished



Organic Chemicals in the Environment

Soaps and Synthetic Surfactants



Detergents A mixture of synthetic surfactants, builders, bleaches, enzymes, and other materials designed to enhance the specific cleaning power

Synthetic surfactants $R–SO_3^-\ Na^+$ **Alkyl sulfonates**

$$[CH_3(CH_2)_nCHOSO_3^-Na^+$$ **Alcohol sulfate**

$$RO(CH_2CH_2CH_2)_nCH_2CH_2OSO_3^-Na^+$$ **Alcohol ether sulfate**

Organic Chemicals in the Environment

Soaps and Synthetic Surfactants

Detergents A mixture of synthetic surfactants, builders, bleaches, enzymes, and other materials designed to enhance the specific cleaning power

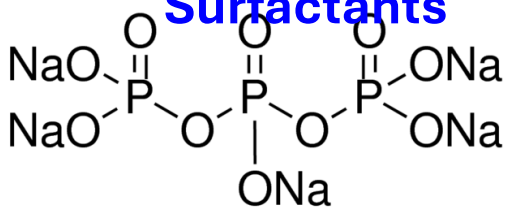
- The main advantage of synthetic surfactants over soaps is that they do not precipitate in hard water

- Builders enhance the cleansing action of synthetics by inactivating the Ca^{2+} and Mg^{2+} in hard water

Ex. polycarboxylic acids, silicates, zeolites (inorganic aluminosilicates) and poly-phosphates

Organic Chemicals in the Environment

Soaps and Synthetic Surfactants



Sodium triphosphate

- Polyphosphates, which were used extensively as builders in the 1970s, were replaced in large part by zeolites because it was believed that phosphate from polyphosphates initiated algal blooms on lakes

Phosphates are nutritious food for alge

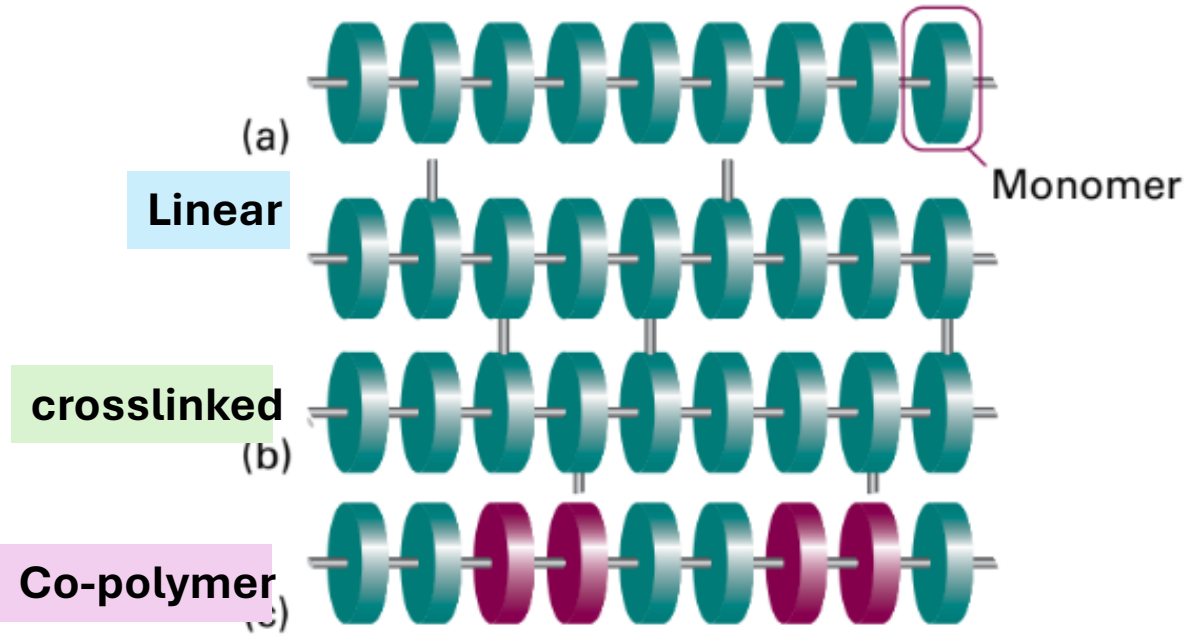
- Detergents account for only 20-25% of the phosphate in lakes and rivers, while the bulk of the remainder comes from fertilizer and animal waste runoff from farms

Seashells get deposited on the beaches and centuries of accumulation of biowaste in some costal areas have turned into rich natural phosphate mines

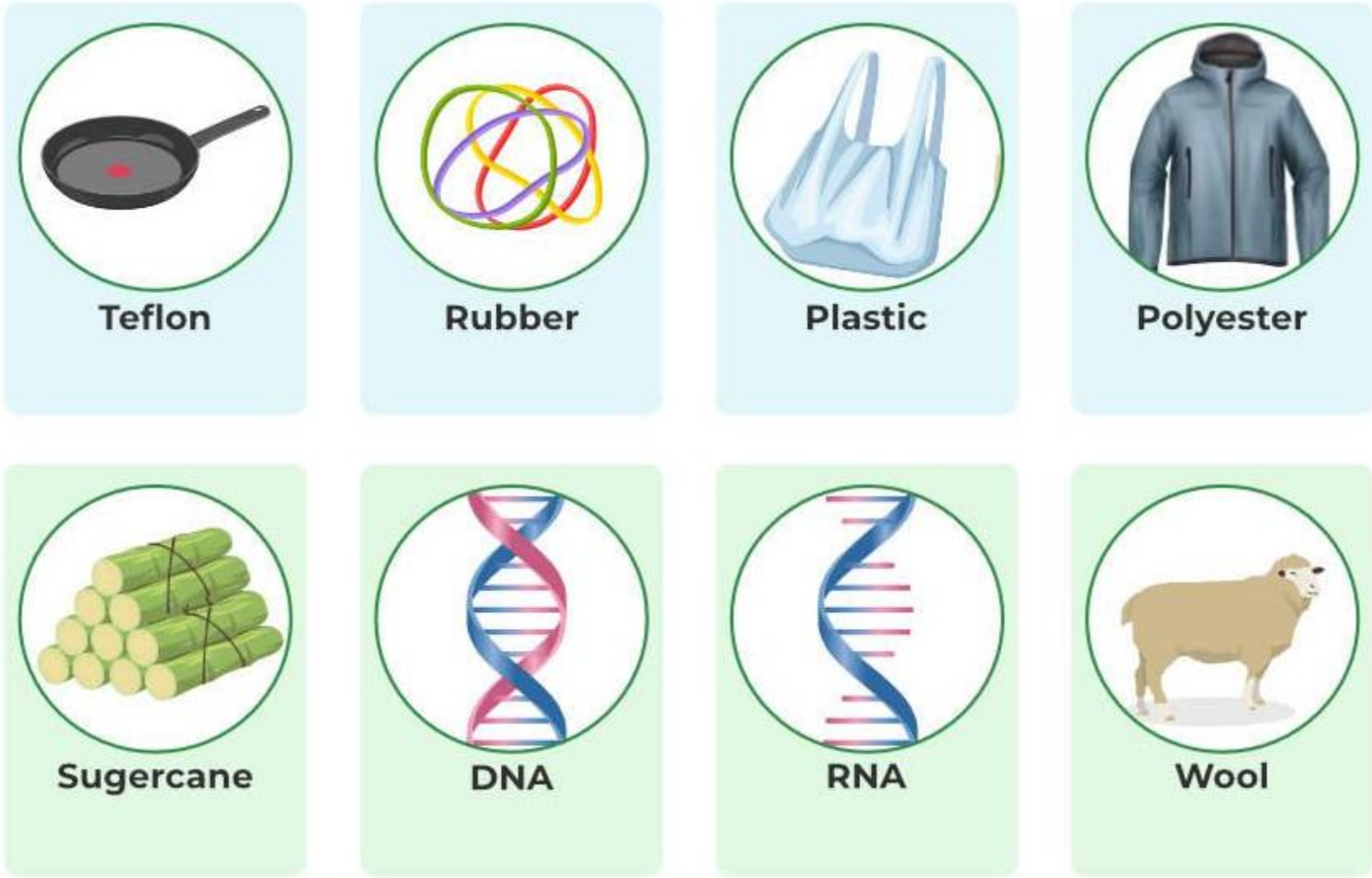
Macromolecules

Macromolecules are very large molecules assembled from smaller molecules
(i) biosynthetically in organisms,
(ii) by chemists in the laboratory, or in an industrial reactor.

Naturally occurring macromolecules include
(i) polysaccharides such as cellulose,
(ii) polypeptides such as protein enzymes, and
(iii) polynucleotides such as deoxyribonucleic acid (DNA).



Which of them are polymers?



ALL

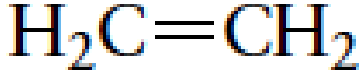
Polymers

- Polymers are also called macromolecules
- They have very large size and high molecular weight (10 kDa to 100 kDa)

Ex.

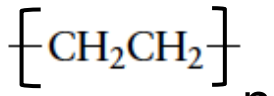
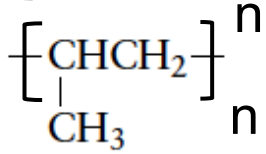
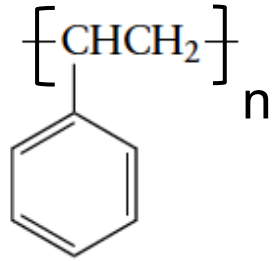
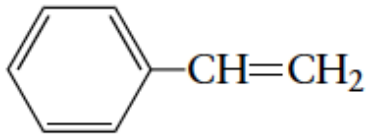
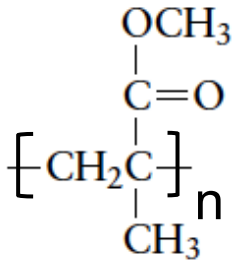
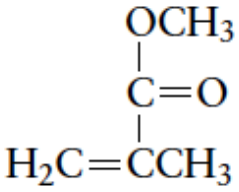
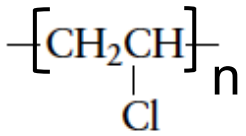
Polyethylene

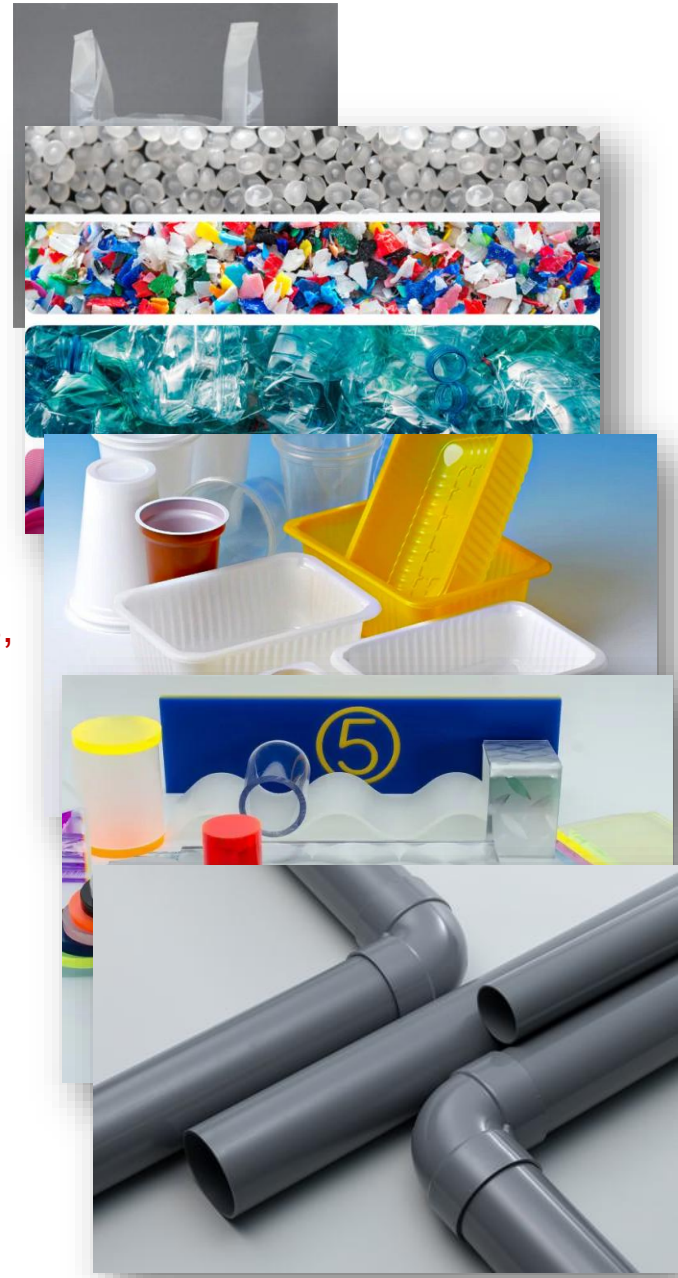
n = degree of polymerization

Synthesized from ethylene 

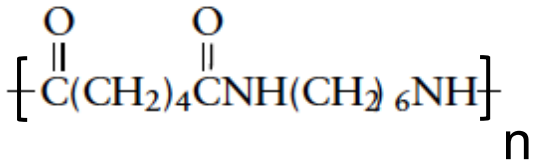
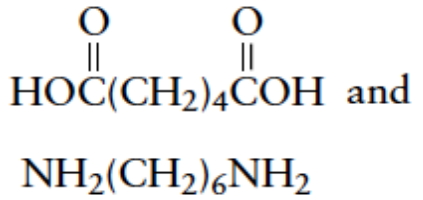
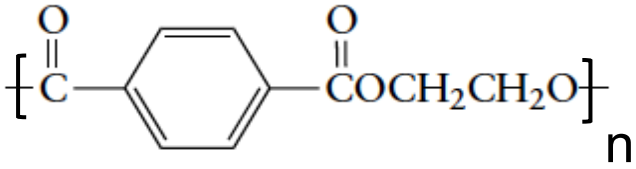
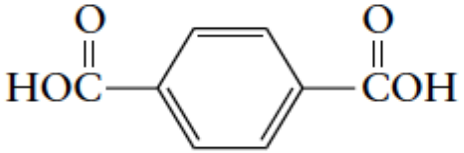
Organic Chemicals in the Environment

Polymers

Polymer	Repeat unit	Monomer(s)	
Polyethylene		$H_2C=CH_2$	Packaging films, bags, bottles and toys
Polypropylene		$CH_3HC=CH_2$	Ropes, tapes, carpets
Polystyrene			Packaging foam, Sterilizing test tubes, lids, bottles
Poly(methyl methacrylate)			Artificial teeth, Bathtubs, Contact lenses
Poly(vinyl chloride)		$H_2C=CHCl$	Window frames, pipes, medical devices



Organic Chemicals in the Environment

Polymer	Repeat unit	Monomer(s)
Nylon 66		 and $NH_2(CH_2)_6NH_2$
Poly(ethylene terephthalate) (PET)		$HOCH_2CH_2OH$ and 

Air bags, carpets, ropes

Water bottles, Containers, Bags

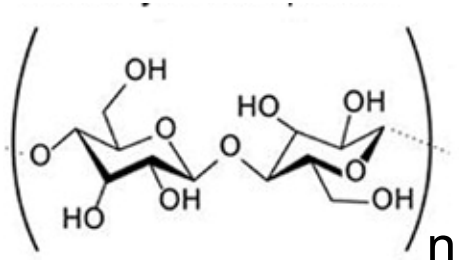
Organic Chemicals in the Environment

Polymers

- Many macromolecules are components of living systems

Proteins: polymers of amino acids

Starch and Cellulose: polymers of sugars



Cellulose

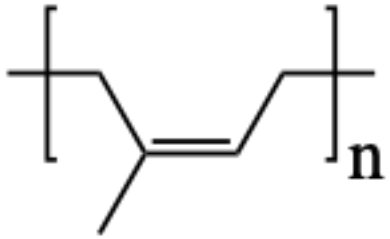
These naturally occurring polymers are biodegradable and will not cause long-term environmental problems

Polymers

In natural rubber 10,000 to 20,000 isoprene units are linked together.

- Natural rubber is not easily biodegradable

Major component



***cis*-1,4- polyisoprene**

Highly resistant to water

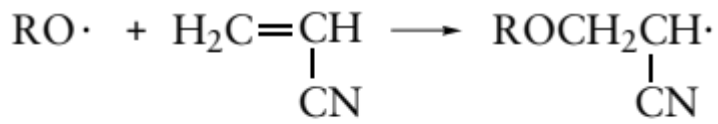
$$n\ CH_2 = \underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}} - CH = CH_2 \xrightarrow{\text{Polymerisation}} \left[CH_2 - \underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}} = CH - CH_2 \right]_n$$

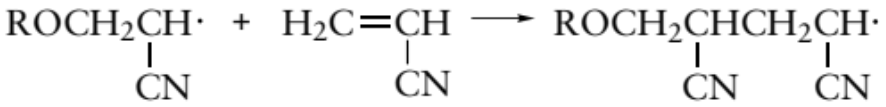
Isoprene
2-methyl–1, 3–butadiene

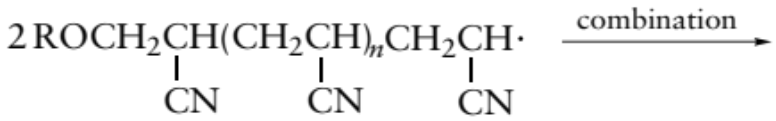
Polyisoprene
(natural rubber)

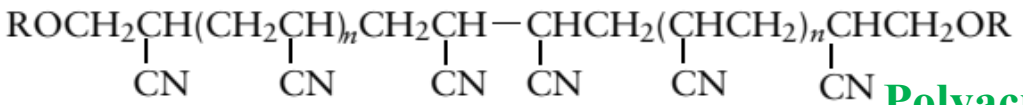
Polymer Synthesis

Chain reaction

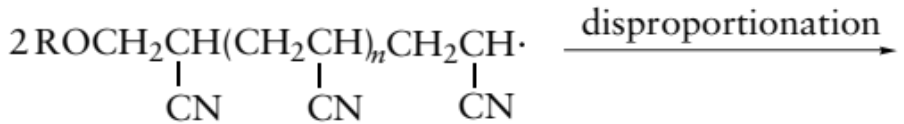


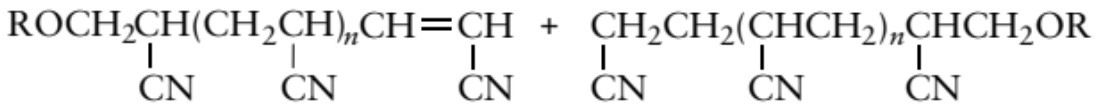






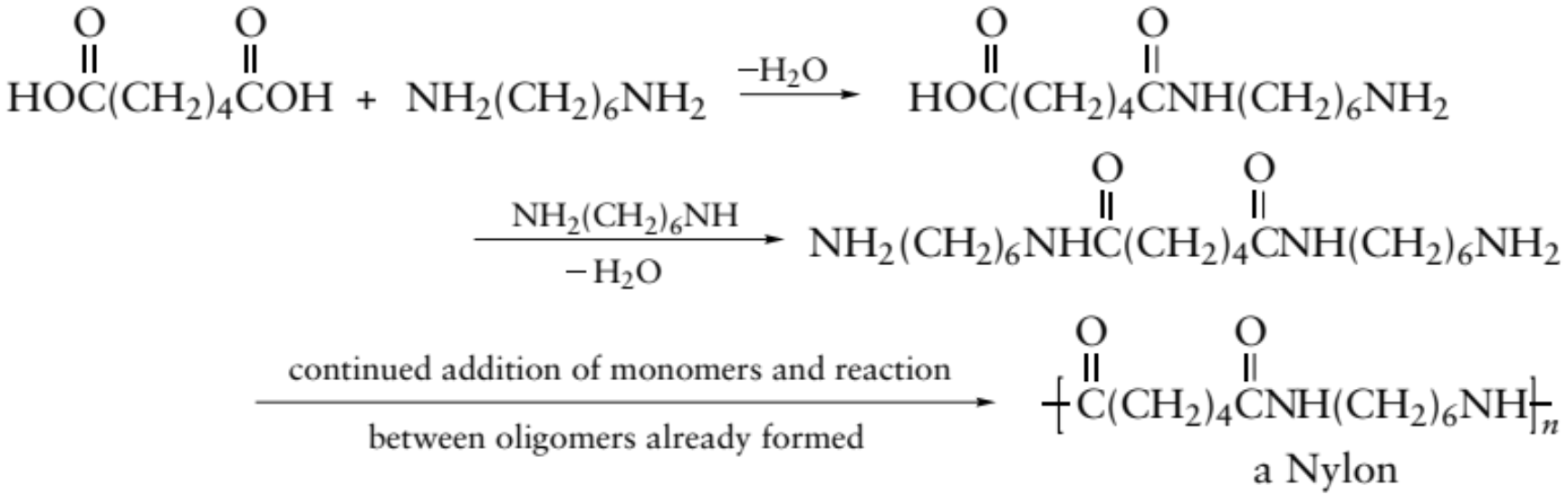
Polyacrylonitrile





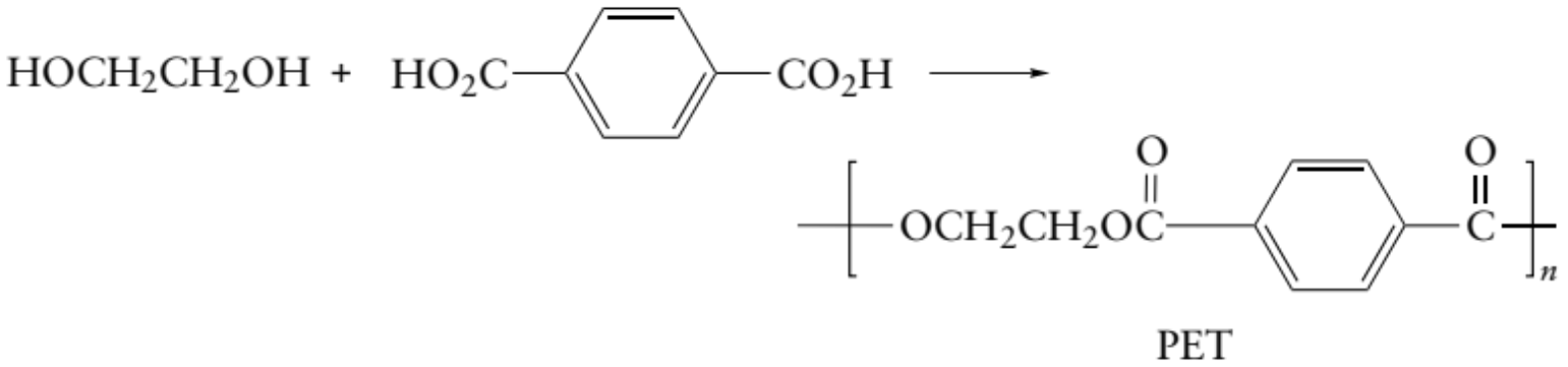
Polymer Synthesis

Step reaction



Polymer Synthesis

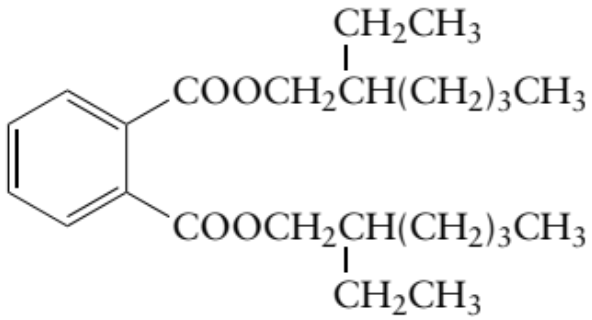
Step reaction



PET

Organic Chemicals in the Environment

Polymer Additives



Plasticizer

Di(2-ethylhexyl)phthalate (DHEP)

- DHEP commonly used as a plasticizer in vinyl polymers
- DHEP can leach from vinyl polymers and cause serious medical problems
- After 2000, use of DHEP as a plasticizer is banned

Plasticizers : Certain plastics do not soften much on heating. These can be easily softened by the addition of some organic compounds which are called plasticizers. For example, polyvinyl chloride (PVC) is very stiff and hard but is made soft by adding di-n-butylphthalate (a plasticizer). Some other common plasticizers are dialkyl phthalates and cresyl phthalate